The Declining Importance of Gender Stereotypes for Politicians' Style in the UK *

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Research on political style suggests that where women make arguments that are more emotional, empathetic, and positive, men use language that is more analytical, aggressive, and complex. However, existing work does not consider how gendered patterns of style vary over time. Focusing on the UK, we argue that pressures for female politicians to conform to stereotypically 'feminine' behavioural styles have diminished in recent years. To test this argument, we describe novel quantitative text analysis approaches for measuring a diverse set of styles at scale in political speech data. Analysing UK parliamentary debates between 1997 and 2019, we document a sharp convergence between men and women for many styles over time. For other styles, we show increasing divergences between genders that run counter to existing stereotypes. An important implication of our findings is that prominent gender-based stereotypes of politicians' behaviour are significantly further from empirical reality now than in the past.

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Introduction

Have incentives for politicians to conform to gender stereotypes diminished over time? In addition to the fact that female and male politicians speak about systematically different sets of political issues (Bäck and Debus, 2018; Catalano, 2009), another dimension on which gendered differences in politics are said to arise is with regard to argumentation style. Gendered communication styles are thought to be rooted in stereotypes that create social expectations for women to act "like women" and men "like men" (Eagly and Wood, 2012). If politicians internalise these expectations before entering politics, or if voters punish them for contravening gender stereotypes (Bauer, 2015, 2019; Boussalis et al., 2020; Cassese and Holman, 2018), legislators will have incentives to engage in gender-role consistent behaviour, and we should expect systematic differences in the political styles that female and male politicians adopt. Empirical evidence supports this view: compared to their male colleagues, female politicians' speeches are more emotional (Dietrich, Hayes and O'Brien, 2019; Gleason, 2019), less complex and jargonistic (Coates, 2015; Lin and Osnabrügge, 2018), less repetitive (Dahlerup, 1988), less aggressive (Grey, 2002; Kathlene, 1994), and use different types of evidence to support their arguments (Hargrave and Langengen, 2020; Mattei, 1998).

However, few studies have considered the degree to which gendered differences in political style might vary over time. This is surprising, as neither the content of stereotypes, nor the social sanctions that are used to enforce them, are necessarily constant (Eagly and Wood, 2012). We build on these insights, and argue that in the context of the UK – where we situate our study – the importance of gender-based stereotypes in politics is likely to have declined over time. We argue that politicians have weaker incentives to adhere to gender stereotypes now than in the past as a result of three factors: changing voter attitudes regarding the appropriateness of gender stereotypes; the increasing

presence of female Members of Parliament (MPs), and of women in leadership positions in parliament; and the increasing homogenisation of politicians on demographic dimensions other than gender. As a consequence we expect to observe a stylistic convergence between men and women in UK politics in recent years.

To evaluate this expectation we examine politicians' styles as they manifest in one prominent legislative activity: parliamentary speech. We conceive of style as a characteristic of speech which is distinct from its content. Intuitively, the style of a speech reflects the *manner* in which an argument is delivered. In social psychology, women's "communal" styles are thought to be marked by higher levels of emotionality, positivity, empathy, and warmth, while men's "agentic" styles are thought to be marked by higher levels of aggression, logic, and confidence (Eagly and Wood, 2012; Schneider and Bos, 2019). Empirical work in political science has operationalised these concepts using a diverse set of indicators. We survey both literatures and identify eight styles that reflect the ideas of communality and agency, and are also – in principle – detectable in the texts of politicians' speeches. The eight styles which we use as the basis of our empirical analysis are human narrative, affect, positive emotion, negative emotion, factual language, aggression, complex language, and repetition.

In addition to our substantive argument that the incentives for politicians to adopt gendered styles have weakened over time, our paper also provides a methodological contribution to the measurement of political style in legislative settings (Boussalis et al., 2020; Dietrich, Hayes and O'Brien, 2019; Proksch et al., 2019). Our goal is to construct measures that closely approximate the conceptual definitions of each of the styles that we highlight in our review of the literature. For some styles, we use existing quantitative text analysis measures that have been extensively validated in other settings (e.g. Kincaid et al., 1975). For others, we develop new measures to detect political style that combine traditional text-as-data dictionary approaches with a word-embedding model.

Our strategy overcomes limitations of standard dictionary approaches as it enables us to detect styles as they manifest *in the specific context of parliamentary debate*. Evidence from a human validation task shows that this approach accurately captures our stylistic categories of interest, and that the measures we develop significantly outperform standard measures that have been used extensively in existing research on political style.

Applying our measures to nearly half a million speeches delivered in the House of Commons between 1997 and 2019, we use a dynamic hierarchical model to describe the evolution of stylistic differences between male and female MPs over time. We report three main findings. First, in the early parts of our study period, we document patterns of political style which are broadly consistent with expectations from the literature on gender stereotypes. Male MPs' speeches are marked by substantially higher levels of aggression and complexity, while female MPs' speeches are considerably more emotional, positive, and make greater use of human narrative. Second, and crucially, we find that these differences have reduced dramatically in recent years. In six out of the eight styles we study, MP gender explains less variation in style use between individuals at the end of our study period than at the beginning. The magnitudes of these changes are large: the proportion of MP-level variation explained by gender decreases by between 29% and 57% from the beginning to the end of our time period, depending on the style. For cases where we report *diverging* behaviour between female and male MPs over time, these stylistic shifts run counter to gender-role expectations. Third, we show that the evolving variation in style use across genders is primarily driven by changes in how women use language over time, rather than the language use of men.

Taken together, our findings suggest that such stereotypes of male and female politicians are substantially less predictive of actual parliamentary behaviour now than they were in the past. Beliefs about how men and women *tend* to behave are often thought to reinforce prescriptive stereotypes about how men and women *ought* to behave (Eagly

and Wood, 2012). As a consequence, how patterns of gendered communication in politics are perceived is important, because such perceptions may confirm biases of voters and the media about the appropriate behaviour of female and male politicians. By documenting the changing use of MPs' styles, we help to root these perceptions in empirical reality, rather than allowing descriptive stereotypes to persist unchallenged. Our focus on elite behaviour contributes to recent work that shows gender stereotypes are of decreasing importance for voters' evaluations of political candidates (Brooks, 2011; de Geus et al., 2020; Saha and Weeks, 2020). We show that, as voter attitudes concerning gender-stereotypes have changed, so too have politicians' adherence to those stereotypes.

Gender, stereotypes, and political style

Why might men and women in parliament employ different political styles? Gender role theory (Eagly and Karau, 2002) suggests that gender stereotypes concerning the typical conduct of men and women can affect behaviour via two main channels. First, repeated exposure to stereotypes from a young age may lead men and women to *internalise* expectations relevant to their genders, which then become self-imposed standards against which they regulate their own behaviour (Eagly and Wood, 2012). Second, descriptive stereotypes (i.e. the perceived tendency for women to be emotional) are often thought to lead to prescriptive stereotypes (i.e. the view that women *should* be emotional), the violation of which leads to the imposition of *social sanctions* by others which further incentivise conformity with gender-based norms (Brescoll and Uhlmann, 2008).

How do these arguments translate to elite women and men? While politicians may also internalise the behavioural stereotypes relevant to their gender, these are likely to be at least partially eclipsed by the expectations associated with their political leadership roles. However, politicians – and particularly female politicians – are nevertheless frequently subject to external pressure for conforming to role-consistent behavioural

standards. First, there is extensive evidence to suggest that voters punish candidates for displaying behaviour that is contrary to gender stereotypes. Voters form gender-biased impressions of candidates (Bauer, 2015, 2019), and penalise women for appearing to be too ambitious (Okimoto and Brescoll, 2010) or negative (Cassese and Holman, 2018), while rewarding them for displays of happiness (Boussalis et al., 2020). As Okimoto and Brescoll (2010, 932) argue, "female politicians are expected to live up to a prescribed level of communality [and] failure to meet those communal standards elicits backlash."

Second, media coverage of candidates for political office is often marked by long-standing gender stereotypes (Kittilson and Fridkin, 2008), and female candidates tend to receive coverage that emphasises deviations from gender-stereotypical norms. Brooks (2013, 1), for example, reports that Hillary Clinton has been criticised in the media for being "angry, aggressive, and unfeminine". Similarly, the characterisation of then UK Prime Minister Theresa May as the emotionless "Maybot" contrasted sharply with gender stereotypes about the supposed emotional sensitivity of women.[1] Taken together, these factors suggest that female politicians face strategic incentives to maintain behaviour that is consistent with stereotypes.

Political scientists have evaluated whether these incentives induce male and female politicians to adopt systematically different political styles. This research focuses overwhelmingly on one aspect of legislative activity where stylistic differences are likely to become manifest: in political speech. These studies can be broadly divided into those that focus on *communal* characteristics of style, which are associated with women, and those that focus on *agentic* characteristics which are associated with men (Eagly and Karau, 2002; Schneider and Bos, 2019). These labels are heuristics for clusters of behavioural attributes, and translating these theoretical concepts into empirically observable features of speech has led researchers to examine an array of diverse styles. In

[1] See "The Making of the Maybot", *Spectator*, 2nd November 2017

surveying this literature, we identified eight distinct styles which prior research associates with stereotypical "masculine" or "feminine" behaviour. We use these styles as the basis of our empirical analysis below.

First, one "communal" characteristic that is typically associated with women is the use of **human narrative**. Women are said to be more likely to use personal experience, narrative, analogies, and anecdotes in their speeches (Blankenship and Robson, 1995; Dahlerup, 1988), an idea that is supported both by the testimonies of politicians themselves (Childs, 2004), as well as in qualitative studies of political speech data (Hargrave and Langengen, 2020). Second, women are also thought to make greater use of emotional language or **affect** (Huddy and Terkildsen, 1993), and there is extensive evidence that women's language exhibits greater overall emotionality than men's (Dietrich, Hayes and O'Brien, 2019; Gleason, 2019; Jones, 2016; Yu, 2013). Third, and more specifically, women appear to be more likely to use **positive emotion** in their political speeches (Boussalis et al., 2020), such as expressing happiness, praise, or satisfaction.

Fourth, as an "agentic" style, men are thought to rely more on **fact-based** language, which is more "analytical, organised and impersonal" (Jamieson, 1988, 76), which relies heavily on empirical evidence, statistics, and figures. In the UK, parliamentarians agree that male politicians pay greater attention to "scientific research" (Childs, 2004, 181), though in other settings there is evidence that women may use more factual language than their male colleagues (Hargrave and Langengen, 2020; Mattei, 1998).

Fifth, male politicians' speech is also thought to feature higher levels of linguistic **complexity**, marked by formalistic and jargonistic word use (Childs, 2004), while women are thought to be more accessible and clear (Coates, 2015; Holmes and Meyerhoff, 2003). Sixth, men are also thought to tend more towards **repetition** in their speeches (Childs, 2004; Dahlerup, 1988). Seventh, men are also said to be more **aggressive**, whereas women are said to avoid a combative and aggressive styles (Brescoll and Uhlmann, 2008;

Karpowitz and Mendelberg, 2014; Kathlene, 1994), and women do appear to be significantly less adversarial and confrontational than men in parliamentary debate (Grey, 2002; Hargrave and Langengen, 2020). Finally, female politicians are also thought to avoid the use of excess **negative emotion** (Cassese and Holman, 2018), while men are said to make greater use of negativity (Brooks, 2011).

Table 1: Political styles

Style	Expected female effect	Definition
Human Narrative	+	Use of personal examples or experiences; the experiences and stories of other people; constituency stories; illustrative examples; referring to individual people, including other MPs.
Affect	+	Use of emotive language in speeches, which might be either positive or negative; such as expressing criticism, praise, disapproval, pride, empathy or fear.
Positive Emotion	+	Use of positive emotional language, which might include expressing empathy, praise, pride, celebration or congratulations.
Fact	-	Use of numbers, statistics, numerical quantifiers, figures and empirical evidence.
Complexity	-	Use of jargonistic, complicated and elaborate language that is challenging to understand.
Repetition	-	Use of the same words or phrases repeatedly.
Aggression	-	Use of aggressive or combative language, which might include criticisms or insults; language that suggests forceful action; or declamatory or adversarial language.
Negative Emotion	-	Use of negative emotional language, which might include expressing fear, anxiety, unpleasantness, sadness or disapproval.

We summarise our eight styles in table 1. For each, we provide a short definition and the expected direction of the gender difference that derives from the literature discussed here.

Evolving incentives for stereotype-consistent behaviour

Despite this rich literature, few studies have examined whether politicians' conformity with gender stereotypes has changed over time. We highlight three mechanisms which we expect to have affected gender-based differences in style in the House of Commons in recent years.

First, public opinion research suggests the *content* of descriptive gender stereotypes has changed over time, as the public perceives women as more agentic and men as more communal now than in the past (Eagly et al., 2020; Sendén et al., 2019). Voters have also become less likely to *endorse* traditional gender stereotypes. For instance, in the context of the UK, 37% of respondents to the British Social Attitudes Survey in 1984 disagreed that "a man's job is to earn money; a woman's job is to look after the home and family", but this disagreement had risen to 72% in 2017 (Taylor and Scott, 2018, 1). More broadly, a growing body of empirical literature also shows that voters rely less on descriptive stereotypes now than in the past (de Geus et al., 2020; Schneider and Bos, 2019), and tend to penalise women only under certain conditions (Brooks, 2011; Krupnikov and Bauer, 2014). Indeed, although contemporary evidence from the US emphasises the importance of gender stereotypes for voter support for women (Bauer, 2015, 2019; Brescoll and Uhlmann, 2008; Cassese and Holman, 2018), recent work from the UK suggests that female politicians do *not* appear to be punished by voters for demonstrating agentic political styles (Saha and Weeks, 2020). To the degree that these findings reflect declining endorsement of gender stereotypes among the UK public, they suggest that the incentives for MPs to conform to stereotypical expectations of behaviour may have also declined in recent years.

Second, the increasing proportion of women elected to political office may also have decreased conformity to stereotypes. The average share of legislative seats occupied by women cross-nationally has increased over time (IPU, 2020). In the House of Com-

mons, women held just 18% of seats in 1997, but this increased to 32% by 2019. The increased descriptive representation of women is important because, as Dasgupta and Asgari (2004, 644) argue, "exposure to counterstereotypic members of a particular social group…can undermine people's automatic stereotypic beliefs about that group." Similarly, Dahlerup (1988, 285) suggests that as the number of women in the legislature increases, "stereotyping decreases, because so many different types of women now occupy the political arena." Consistent with these claims, higher levels of female representation have been shown to reduce public endorsement of gender-based stereotypes (Kerevel and Atkeson, 2015). Moreover, female politicians in the UK occupy more high-powered positions within the legislative hierarchy now than in the past (Uberoi, Watson and Kirk-Wade, 2020). The increasing prevalence of women in leadership positions reduces the degree to which communal qualities are ascribed to women and leadership qualities to men (Dasgupta and Asgari, 2004). If gender stereotypes are weakened when the number and profile of women in parliament increases, we should expect stylistic differences between men and women that are relevant to those stereotypes to decrease as well.

A wealth of empirical evidence suggests that many behavioural differences between men and women in political settings diminish when there are more women present. As the proportion of women grows, participation becomes more equitable (Karpowitz and Mendelberg, 2014), women become more influential (Karpowitz, Mendelberg and Shaker, 2012), are interrupted less (Mendelberg and Karpowitz, 2016), and network more (Holman and Mahoney, 2018). The presence of female leaders also reduces gender gaps in participation and influence in parliamentary debate (Blumenau, 2019). Together, these findings suggest that increasing numbers of women in high-profile positions may depress general gendered behavioural differences – something that we expect to also apply to debating styles.

By contrast, an alternative view suggests that when women hold only few "tokenised" positions, they are likely to be penalised for behaviour that contravenes *organisational* role stereotypes which are primarily defined by typical male behaviours. When women operate in male-dominated settings, they are incentivised to "blend unnoticeably into the predominantly male culture" Kanter (2008, 973) and may have to "capitulate most of their … femininity" in order to succeed (Childs, 2004, 187). The implication of these arguments is that gender differences in style will be smallest in early years when all politicians use "masculine" styles, but larger differences between genders will develop as more women enter the legislature. While this perspective suggests gender gaps will have *increased* in recent years, it nevertheless engages with the idea that stylistic differences between men and women are unlikely to be constant over time. Our empirical analysis allows us to test these competing perspectives about the effects of the increased descriptive representation of women on gendered differences in style.

Third, changes in style might also result from cohort changes among the types of politician elected. In the UK, politicians suggest that those elected in later cohorts may have encountered fewer experiences that are "determined by traditional bifurcated gender roles" (Childs, 2004, 185). Similarly, if politicians are becoming more homogeneous on other demographic dimensions, this may also reduce the importance of gender for determining behaviour. Candidates for political office in the UK are increasingly likely to be middle-class, university educated and to have white-collar backgrounds (Lamprinakou et al., 2017; O'Grady, 2019). To the degree that such features are also predictive of MPs' styles, the increasing homogeneity of politicians on these dimensions may work to suppress existing gender-based behavioural differences.

Taken together, these arguments suggest that behavioural differences that are relevant to gender stereotypes are likely to have decreased in the UK in recent years. Few prior studies have evaluated changes over time in gender gaps in style, and those that

do exist are relatively narrow in focus. Jones (2016) studies the style of a single individual – Hillary Clinton – and shows that she became more "masculine" over time as her political power increased. Grey (2002) shows the increased presence of female MPs in New Zealand led other women to use increasingly aggressive language. By contrast, Yu (2013) finds that US Congresswomen's speeches contained both "feminine" and "masculine" language at levels that remained approximately constant over nearly 20 years. We build on these works by adding evidence from an additional legislative setting, and also by significantly expanding the set of styles studied.

Data and Methodology

We consider the words of politicians' speeches as the primary locus of political style in parliament and the strategies we outline below each use texts of political speeches to infer the styles adopted by different speakers. Parliamentary speech is a useful source of information for measuring political style as it provides long-running panel data at the individual level. In addition, legislative debates occur under specific rules which change only rarely over time. In the UK, debates operate under relatively permissive rules: MPs are afforded a large degree of autonomy regarding the debates to which they contribute, and party leaders exert no control over who participates, nor over the content of speeches that MPs deliver.

We study House of Commons debates between May 1997 and March 2019. Our choice of study period is motivated by the fact that prior to the 1997 election women accounted for less than 10% of all MPs in the Commons, and so analysis of earlier periods would likely be sensitive to the styles of the specific women included in the analysis. We collapse our data such that all speeches made by an MP in a debate constitute a single speech-document, making our unit of analysis an individual MP in a debate. We remove all speech-documents shorter than 50 words, as well as contributions by the Speaker of

the House, whose speeches are almost entirely procedural. We also exclude any debate that has fewer than five participants.[2] Our final sample consists of 14,864 debates, 1,422 MPs (370 female, 1,052 male), and 418,147 MP-debate observations.

Measuring "style" with context-specific dictionaries

A common approach to measuring latent concepts, such as style, in text data is to assign each text a score on the basis of a predefined dictionary that aims to capture the concept of interest. However, dictionary-based approaches are highly domain-specific, as the words used to capture a concept in one context – say, parliamentary speeches – are likely to be different to those used to express the same concept in another context. As Grimmer and Stewart (2013, 268) argue, "applying dictionaries outside the domain for which they were developed can lead to serious errors." This critique is of particular concern for previous studies of gendered styles, many of which apply "off-the-shelf" dictionaries to various different political contexts (e.g. Gleason, 2019; Jones, 2016; Yu, 2013).

We propose an alternative approach that combines standard dictionaries with a locally-trained word-embedding model to construct domain-specific dictionaries that are better able to capture our styles of interest *as they manifest in the context of UK parliamentary debate*. The key advantage of our approach is that it allows us to account for context-specific patterns of word use. That is, rather than simply using an off-the-shelf dictionary that may be poorly suited to capturing, for instance, aggression in the parliamentary setting, this approach allows us to automatically create a bespoke aggression dictionary which is firmly rooted in the way that vocabulary is used in parliamentary debate. We use this approach to measure six of our styles: aggression, affect, positive and

[2]We make this exclusion because our model can become computationally burdensome with very large numbers of debates. Small debates attended by few MPs contribute little to our estimates given the random-effect structure of the model described below, and so our results are very unlikely to be sensitive to this decision.

negative emotion, fact, and human narrative.

For each of these styles, we follow three steps to construct the relevant score for each speech in our corpus. First, we define a "seed" dictionary for each style that represents our concept of interest. For four of our styles (affect, negative emotion, positive emotion, and fact), we use existing dictionaries and for two of our styles (aggression and human narrative) we create our own seed dictionaries based on a close reading of a sample of parliamentary texts.[3]

Second, we estimate a set of word-embeddings using the GloVe model described by Pennington, Socher and Manning (2014). Word-embedding models rely on the idea that words which are used in similar contexts will have similar meanings, and the embedding model allows us to *learn* the semantic meaning of each word directly from how the word is used by MPs in debate. We train the embedding model on the full set of parliamentary speeches, and the main output of the model is the set of word-embeddings themselves. These are dense vectors that correspond to each unique word in the corpus, the dimensions of which capture the semantic "meanings" of the words. Crucially for our purposes, given this representation, the distances between word-vectors have been shown to effectively capture important semantic similarities between different words (Mikolov et al., 2013).

By calculating the cosine similarity between every word in the corpus and the words in each of our seed dictionaries, we can therefore use this property to define the set of words that, *in the specific context of parliamentary debate*, are used in a semantically similar fashion to the seed words. We label this quantity as Sim_w^s, where w indexes words, and s indexes each of the style dimensions. Words closely related to the average semantic meaning of the seed words for a given dictionary will have a high similarity score (close to 1), and words that are less closely related will have a low similarity score

[3]We include a full description of the seed dictionaries in the appendix.

(close to 0). We provide full details of our approach in the appendix.

The Sim_w^s scores therefore define a domain-specific dictionary for a given style type. They describe the degree to which each unique word in our corpus is used similarly to the ways in which the words in our seed dictionary are used, on average. For instance, if our seed dictionary for aggression included the words *disgraceful* and *outrageous*, we might discover that a word that is absent from the seed dictionary – say, *dreadful* – was also frequently used in similar contexts in other parliamentary speeches. If this is true for many instances of the word *dreadful*, then that word would receive a high Sim_w^s score as the result of our word-embedding approach. In essence, the embeddings enable our seed dictionaries to automatically expand to incorporate words that are used in a similar manner to the words that they already include. In the appendix we show that our approach is very successful at "adding" words that are representative of our styles but do not appear in our seed dictionaries (including "dreadful" to the aggression dictionary). This approach also "removes" words that appear in our seed dictionaries but are clearly not representative of our styles in the parliamentary setting. For instance, the word "terrorist", although included in our seed dictionary for negative emotion, is given a very low Sim_w^s score as that word is typically used to refer to matters of policy in parliament, rather than being used as an expression of emotion. Standard dictionary approaches would miss these features of language use and, as we show in our validation, would result in considerably weaker detection of the use of parliamentary styles.

Third, we use the word-level scores, Sim_w^s, to score each *sentence* in the corpus on each style dimension according to the words they contain. In particular, the score for a given sentence on a given dimension is:

$$Score_i^s = \frac{\sum_w^W Sim_w^s N_{wi}}{\sum_w^W N_{wi}} \tag{1}$$

where Sim_w^s is the similarity score defined above, and N_{wi} is the (weighted) number of

times that word w appears in sentence i, where the weights are term-frequency inverse-document-frequency weights.[4] When words with high scores for a given style appear frequently in a given sentence, the sentence will be scored as highly relevant to the style. The score for each *document* is then the weighted average of the relevant sentence level scores, where the weights are equal to the number of words in each sentence.

The approach we outline here is similar to that developed in Rice and Zorn (2019), who also use a word-embedding model to create context-specific dictionaries. We provide two important extensions to their work. First, the approach described by Rice and Zorn allows researchers to locate documents on a domain-specific, positive-negative sentiment dimension. While sentiment scores are important in some contexts, our approach allows researchers to score documents for *any* arbitrary latent concept that they define by constructing a small seed dictionary. Second, Rice and Zorn apply their method to US Supreme Court opinions, but suggest that a key concern is whether word-embedding dictionary construction "[yields] valid dictionaries for widely-varying types of specialised vocabularies" (Rice and Zorn, 2019, 15). We extend the idea of word-embedding based dictionaries to a new setting – the UK House of Commons – and to six new specialised vocabularies which relate to the literature on gender stereotypes. As our validation exercises show, there is strong evidence that this approach significantly outperforms standard dictionary approaches across the set of styles we study.

Measuring "complexity" and "repetition"

For our final two styles – complexity and repetition – dictionary approaches (domain-specific or otherwise) are unsuitable, as these styles are not detectable from the occurrence of specific words. Instead, we adopt two different metrics to capture these concepts. For *complexity*, we use the Flesch-Kincaid Readability Score (Kincaid et al.,

[4] TF-IDF weighting is used to down-weight very common words, and up-weight relatively rare words.

1975). The intuition behind this measure is that documents that have fewer words per sentences, and fewer syllables per word, are easier to understand (more "readable"). In addition to providing general validation, Benoit, Munger and Spirling (2019) show that this metric correlates very highly with other similar measures. We multiply the original formulation of this score by -1 such that higher numbers indicate higher levels of text complexity.

We consider MPs to be *repetitive* when they use the same language repeatedly during a debate. To measure repetition we use the algorithm introduced by Ziv and Lempel (1977), which is a lossless text-compression algorithm that underpins a variety of common computer applications. Compression algorithms work by finding repeated sequences of text and using those patterns to reduce the overall size of the input document. The efficiency of the compression of a text is directly related to the number and length of the repeated sections in that text. We apply the compression algorithm to every document in our corpus, measure the degree of compression, and treat that quantity as the measure of repetition for each MP in each debate. Simply put, the more compression that a speech receives, the more repetitive we deem it to be.

Finally, to put our eight style measures on comparable scales, we normalise each measure across documents to have mean zero and standard deviation one. This means that average differences for each style between men and women can be interpreted in standard deviations of the outcome variable.

Validation

Table 2 provides results from a human validation task which assesses whether our text-based measures of style mirror human judgements of the same concepts. Coders were presented with randomly-sampled pairs of sentences and were asked to complete two tasks. First, a style-*comparison* task required respondents to select which of the two

Table 2: Correlation between text-based measures and human judgments.

Style type	Comparison task	Intensity task
Human Narrative	0.71 (0.43)	0.69 (0.44)
Affect	0.66 (0.49)	0.64 (0.48)
Positive Emotion	0.71 (0.32)	0.71 (0.39)
Negative Emotion	0.74 (0.46)	0.77 (0.48)
Fact	0.81 (0.74)	0.71 (0.7)
Aggression	0.72 (0.21)	0.66 (0.34)
Complexity	0.83	0.82
Repetition	0.64	0.77

sentences included language that was more typical of a particular style. Second, a style-*intensity* task required respondents to rate the degree to which each sentence was representative of the selected style on a 5-point scale. Our coders collectively completed approximately 140 sentence comparisons per style (280 sentence ratings).[5] We provide more details on this task, as well as a series of further validity checks, in the appendix.

We summarise the results in table 2. The "intensity task" column presents the correlation between our sentence-level style measures (equation 1) and our coders' ratings of the same styles. For the "comparison task" column, we calculate the difference in the sentence-level scores for each pair of sentences, and correlate that with the choices made by our coders from the comparison task. For the styles that can also be quantified using standard dictionary measures, the numbers in parentheses show equivalent correlations between those measures and the judgments of our coders.

Overall, the results are very encouraging. Across all styles, the correlation between the text-based scores and the human validation is always positive, and is never lower than 0.64 for either task. These results suggest that there is a clear correspondence

[5]To assess inter-coder reliability, our research assistants both coded an additional common set of 20 comparisons per style. Coders agreed on which of the two sentences was more representative of a given style in 75% of comparisons. The correlation for the "intensity" scores for all sentences across coders was 0.8.

between the measures of style implied by our text-analysis approach, and human judgments of those concepts in the same set of texts. Further, our word-embedding approach clearly outperforms standard dictionary approaches in approximating human judgment. For instance, for positive emotion, standard dictionary measures correlate at 0.32 and 0.39 with human codings for the two tasks, compared to 0.71 and 0.71 for the word-embedding approach. Despite the relatively small sample sizes for each style, the magnitude of the difference in predictive power means that – in all cases except for "fact" – the correlation between our word-embedding measures and human codings is significantly higher than the equivalent correlation for standard dictionary measures. This exercise provides strong evidence that we are able to reliably detect our styles of interest in parliamentary speech and outperform the standard measures used in previous studies on gender and political style.

Modelling political style

Our goal is to assess the degree to which style use varies by MP gender, and whether such differences change over time. There are two main inferential concerns for establishing these patterns. First, we might worry that female and male legislators are not alike, particularly with regard to institutional characteristics like tenure in parliament and seniority, and that these differences might also be correlated with language styles. If confounders are present, then changes we observe in gendered use of particular styles over time might be attributable to these alternative factors. Second, men and women systematically participate in debates devoted to different topics (Catalano, 2009), and there is evidence that debate topic also correlates with style (Osnabrügge, Hobolt and Rodon, 2020), and we may worry that these patterns will work to confound our inferences. For instance, if women participate more in debates on education which contain language related to human narrative, while men participate more in debates on the

economy which include more factual language, then gender differences in topic usage will confound gender differences in style.

To account for these concerns, we adopt a Bayesian dynamic hierarchical model that allows us to account for a wide variety of both individual- and topic-level confounders, while also flexibly estimating changing gender dynamics in style over time. For each speech i, we have a continuous measurement of style s, which we denote as y_i^s. For speech i, by MP j, in debate d, and time period t, we model the data as a function of individual- and debate-level parameters:

$$y_{i(jdt)}^s \sim N(\alpha_{j,t} + \delta_d, \sigma_y) \tag{2}$$

where $\alpha_{j,t}$ is an MP-specific random effect which captures average differences in MP style use. The t subscript indicates that we fit one intercept for each MP in each time period that they appear in the data, thus allowing us to capture average style use at different points in time. We use parliamentary sessions as our unit of time. These sessions normally last for about one calender year, and there were 20 sessions between 1997 and 2019. We observe speeches from 635 MPs on average in each session, and each MP appears in 9 sessions on average. The δ_d parameters are random effects which capture average differences in style use in different debates.

Our primary interest is in describing variation in the $\alpha_{j,t}$ parameters. We model these random-effects at the second level of the model as a function of MP gender, while allowing the relationship between gender and style-use to vary over time:

$$\alpha_{j,t} \sim N(\mu_{0,t} + \mu_{1,t} Female_j, \sigma_\alpha) \tag{3}$$

Here, $\mu_{0,t}$ represents the average use of a given style among male MPs in time period t, and $\mu_{1,t}$ describes the average difference in style use for women relative to men,

again in time period t. The standard deviation σ_α, describes how much, on average, the MP-session intercepts vary around the mean style use for MPs of each gender.[6] It is unrealistic to assume that gender differences in one parliamentary session are independent of those in previous sessions, and in order to reflect a more realistic evolution of these differences we model the $\mu_{0,t}$ and $\mu_{1,t}$ parameters as a first-order random-walk process:

$$\begin{aligned} \mu_{0,t} &\sim N(\mu_{0,t-1}, \sigma_{\mu_0}) \\ \mu_{1,t} &\sim N(\mu_{1,t-1}, \sigma_{\mu_1}) \end{aligned} \tag{4}$$

This specification assumes, *a priori*, that the average use of a style by women and men will be similar in t and $t+1$, and that changes over time will therefore occur gradually. This encourages smooth coefficient changes over time, but still allows for large deviations from one period to the next if the information from the data is sufficiently strong.

The $\mu_{0,t}$ and $\mu_{1,t}$ parameters are our main quantities of interest. $\mu_{1,t}$ captures the difference in average style use between genders in each time period, and our review of the theoretical literature implies general expectations for the *sign* of $\mu_{1,t}$ for each style (see table 1). Consistent with our theoretical discussion of how the incentives for conforming with gender stereotypes have changed in recent years, we also expect the *magnitude* of $\mu_{1,t}$ for each style to decrease over time. We therefore also calculate R^2-style statistics which describe the proportion of individual variation in style use explained by MP gender in each parliamentary session.[7] We report both quantities below.

[6] Note that we use a common variance parameter for all time periods.

[7] We follow Gelman and Pardoe (2006) and calculate

$$R^2_{\alpha,t} = 1 - \frac{E(V_{j=1}^{J} \hat{\epsilon}_{j,t})}{E(V_{j=1}^{J} \hat{\alpha}_{j,t})}$$

where

$$\hat{\epsilon}_{j,t} = \hat{\alpha}_{j,t} - \hat{\mu}_{0,t} + \hat{\mu}_{1,t} Female_j$$

Note that this model structure allows us to account for the two sources of confounding that we describe above. We can account for individual-level confounders by including a set of MP-specific covariates into the model. To do so, we replace equation 3 with:

$$\alpha_{j,t} \sim N(\mu_{0,t} + \mu_{1,t} Female_j + \sum_{k=1}^{k} \lambda_k X_{j,t}^k, \sigma_\alpha) \tag{5}$$

where $X_{j,t}$ is a vector of individual-level covariates which can vary by session. There are three sources of individual-level confounding that are of particular concern. First, MPs in leadership positions may use systematically different debating styles than backbench MPs. Women have come to occupy a greater share of legislative leadership roles over the period we study (Blumenau, 2019), meaning that changes in style use between male and female MPs over time might be attributable to changes in the people who occupy these roles. We therefore control for whether the MP held a frontbench position for either the government or opposition in a given session, and also for whether they were a committee chair.

Second, the Labour Party has historically had significantly more female MPs than other parties in the Commons, though the partisan gender gap has diminished to some degree over time (Uberoi, Watson and Kirk-Wade, 2020). If MPs from different parties use styles at different rates, then any change we observe in gendered use of particular styles might be confounded by the fact that proportionally more Conservative Party female MPs have been elected to parliament in recent years. We therefore also include a set of party dummies in the specification for $X_{j,t}$.

Third, even conditional on party, opposition MPs use significantly more negative language than government MPs (Proksch et al., 2019). We might therefore be concerned that average shifts in the gendered use of certain styles merely reflect the movement of female MPs into and out of government. In our context, the Labour Party was in government between 1997 and 2010, but moved into opposition after the 2010 election. As

a result, because Labour has proportionally more women, we might observe an average increase in women's use of more agentic styles simply as a function of changes in MPs' government and opposition status. To address this concern, we add a dummy variable to $X_{j,t}$ which captures whether the MP is a member of a governing or an opposition party. Finally, we also add individual-level controls for MP age and years served in parliament.

In addition, the debate-level intercepts, δ_d, mean that it is only *within-debate* variation in male and female style use that informs the estimates of our central quantities of interest. In other words, $\mu_{1,t}$ will capture only the degree to which men and women use different styles when speaking about the same substantive topic. Similarly, changes in $\mu_{1,t}$ over time will also be free from debate-level topical confounding.[8]

We estimate our model separately for each style in Stan (Carpenter et al., 2017), where we use three chains of 500 iterations, after 250 iterations of burn-in.

Results

Figure 1 shows the values of $\mu_{1,t}$ – the average difference between men and women for each style type, in each parliamentary session. Positive values indicate that a style is used more by women, and negative values indicate higher use of the style by male MPs. The green shading indicates the expected direction of the gender effects based on previous literature (see table 1).

Given these expectations, we find – in the early years of our sample – broad sup-

[8]Debate intercepts are drawn from a mean-zero normal distribution, with estimated variance:

$$\delta_d \sim N(0, \sigma_\delta) \tag{6}$$

Our model is completed by normal prior distributions over the λ parameters:

$$\lambda^k \sim N(0, 2) \tag{7}$$

and half-normal prior distributions over the variance-parameters:

$$\sigma_\alpha, \sigma_{\mu_0}, \sigma_{\mu_1}, \sigma_\delta, \sigma_y \sim N(0, 2) \tag{8}$$

port for the hypothesis that male and female political styles will conform to gender stereotypes. For instance, female MPs are indeed more likely to draw on examples that emphasise human narrative, and are also more likely to use positive and emotional language. Similarly, we find support for the idea that men will use more aggressive and complex language, at least before 2010. By contrast, for negative emotion, fact, and repetition, our findings contrast with expectations: in each case, female MPs are more likely to express these styles. Overall, in the first half of our sample, this evidence is broadly consistent with the expectations of the literature on gender stereotypes.

However, figure 1 also reveals that there is significant variation in the size of these gender differences over time. For affect, positive emotion, and human narrative, women are more likely to use these styles in the early period in our data, but gender differences become much smaller over time. For positive emotion and affect, there is no consistent significant difference between men and women by the latest years in our data. Similarly, while men are significantly more aggressive than women before 2010, this difference has also disappeared in recent years. Consistent with gender stereotypes, men use more complex language in earlier periods, but we see a sharp convergence in the complexity of men and women starting around 2011. For repetition, women are somewhat *more* repetitive throughout our sample, but there has also been a gradual convergence between men and women in the use of this style.

Further, for some of our styles we observe increasing differences between genders over time, but the direction of these shifts run *counter* to stereotypes. This is most notable for fact and negative emotion, which are representative of agentic male communication styles. Though there are negligible gender differences in the earlier period, from 2007 onward, women use significantly more factual language than men. Similarly, although women use negative emotion in their speeches at higher rates throughout the time period, this gender difference has grown substantially larger over time.

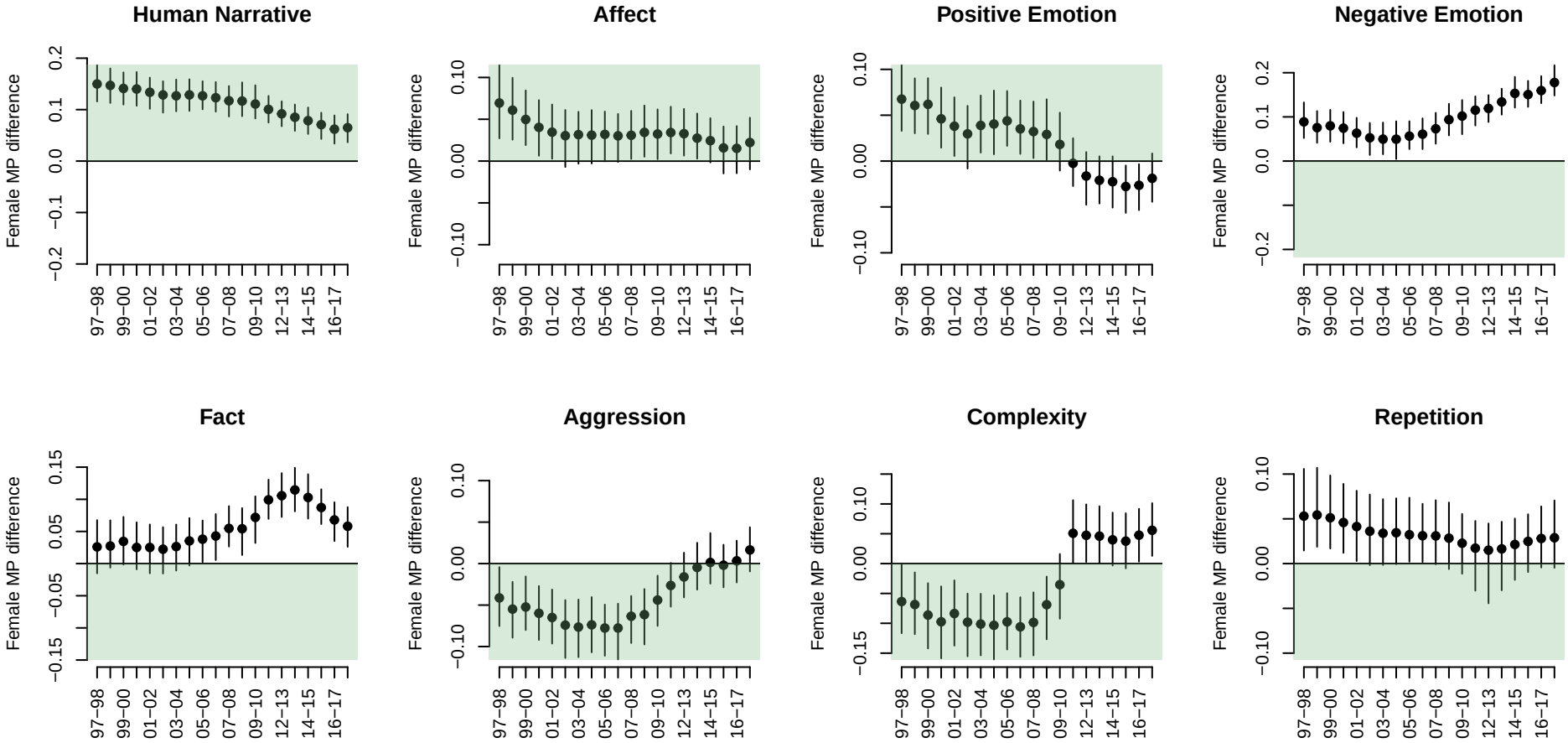

Figure 1: Gender differences in style over time

Using the R^2_α statistic defined above, we can quantify the degree of individual variation in style use which is explained by MP gender over time. For six styles, gender explains significantly less variation in the second half than in the first half of our study period. For instance, between 1997 and 2007, gender accounts for, on average, 6.4% of the MP-level variation in the use of human narrative, but this decreases to just 3.5% on average for the period between 2007 and 2019 – a decrease of nearly 50%. Similarly, the average proportion of variation in style use explained by gender decreases by approximately 45% for affect; by 38% for positive emotion; by 29% for aggression; by 57% for complexity; and by 56% for repetition. For fact and negative emotion, gender explains a larger proportion of the variation in style use later in the study period but, again, these changes run in the opposite direction to the expectations of the literature on gender stereotypes.

Are these patterns we present above the result of changes in the behaviour or male or female MPs? In figure 2, we depict changes in style use over time separately for women and men. The plot reveals that for most styles, the more significant shifts have been among women. For instance, while there has been an overall decrease in use of human narrative, this decrease is more pronounced for women than it is for men. Similarly, while men's use of negative and factual language has remained fairly stable over time, there has been a sharp increase in women's use of these styles in recent years. Across almost all the styles we study, the largest year-to-year shifts in speechmaking behaviour occur among women. In general, the speech styles of female MPs in the UK are markedly more agentic and less communal now than in the past, implying that the descriptive validity of prominent gender-based stereotypes of political behaviour is in decline.

In the appendix, we investigate whether gender explains less variation in aggregate style use because of a gradual convergence in styles of men and women throughout their careers ("within-MP" effects), or because the men and women entering parliament

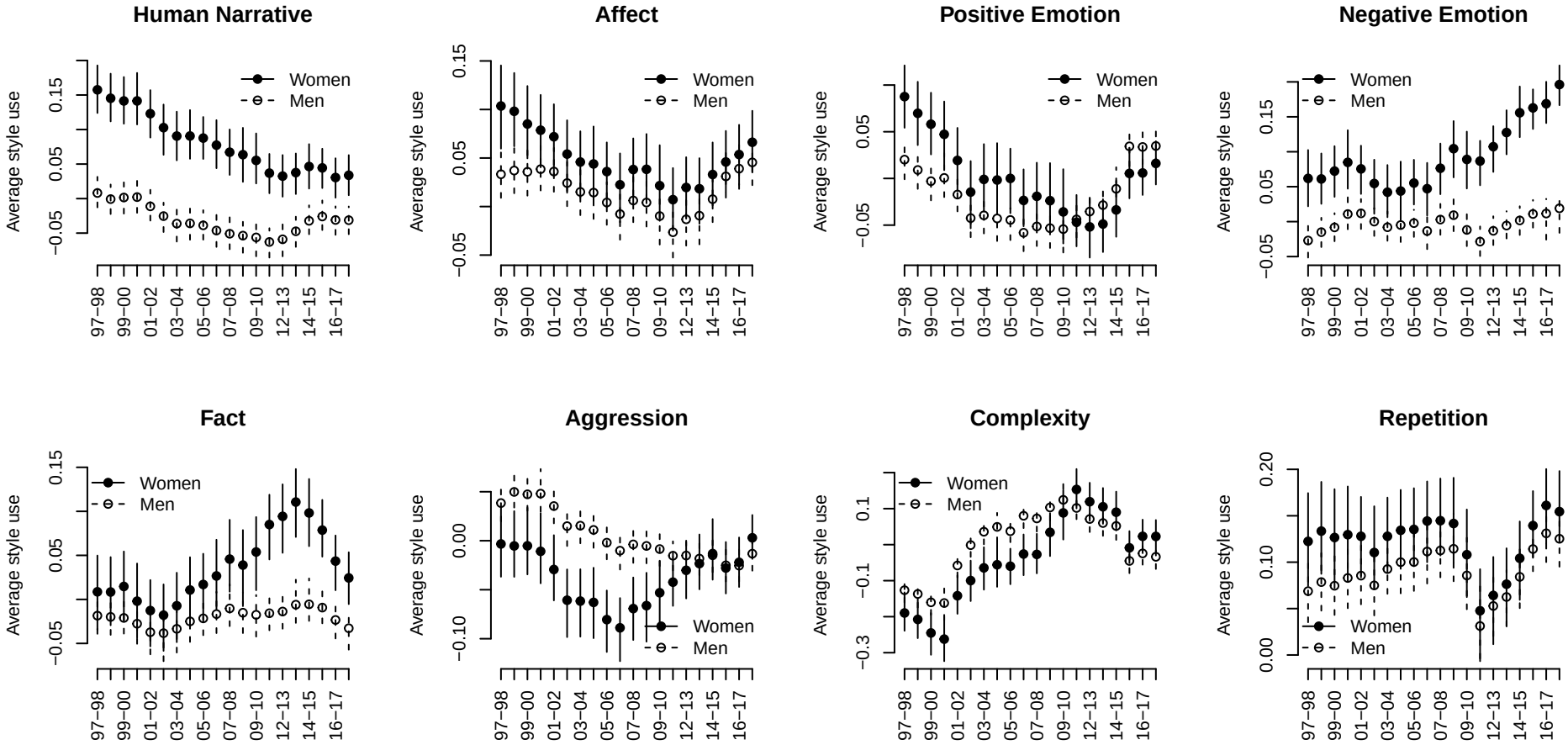

Figure 2: Average style use for men and women over time

over time are systematically different from those leaving parliament ("replacement" effects). While there is some evidence that replacement appears to be more important for explaining the changes in "agentic" styles and within-MP change is somewhat more important for explaining change in "communal" styles, overall we find that neither replacement nor within-MP change alone explain the convergence that we document above.

We have argued that the incentives for politicians to conform to traditional stereotypes have diminished over time, but there are alternative explanations that could account for these patterns of behaviour. We address a number of these concerns here. First, in the appendix we implement the model described in equation 5 and control for MP-session covariates, including party, opposition status, and institutional position. If the changes we observe over time are driven by these factors, we would expect to see large differences between these two sets of results. We do not. Although there is some attenuation of the over-time changes for aggression and complexity when controlling for covariates, in general we observe stereotype-consistent differences at the beginning of the sample period but a clear convergence between men and women in recent years. Importantly, the over-time patterns we document are not attributable to the fact that the Labour Party, which has the majority of women MPs, became the party of opposition in 2010.

Second, the aggregate patterns we document here could reflect changes to the parliamentary agenda, rather than changes in behaviour from male and female MPs. Imagine, for instance, that women are more likely to use human narrative than men when discussing a particular topic, such as schooling. If schooling is frequently discussed in parliament early in our data, but only infrequently in the latter period, then average gender differences in human narrative would decrease even in the absence of a change in behaviour of either male or female MPs. In other words, our results might be confounded by changes in the prevalence of different topics over time. In the appendix we use sta-

tistical topic-models to measure the average differences between male and female style use across a wide variety of topics, and then evaluate whether topics that are marked by large stylistic differences become more or less prevalent over time relative to topics marked by smaller differences. We find scant evidence of such topical confounding.

Conclusion

Our central substantive contribution is to document the fact that gender stereotypes are worse descriptors for actual political behaviour in the UK now than was true in the past. In particular, in recent years, women in the House of Commons demonstrate less "communal" and more "agentic" political styles, and the gender gap on most dimensions of style that we examine has decreased. We see these results as an important corrective to the scholarly literature on gender differences in legislative behaviour, which typically emphasises that male and female politicians argue in ways that are broadly consistent with stereotypes. Though this may be true in some settings, gender stereotypes of communication styles have become significantly less predictive of the reality of contemporary British political debate.

We focus only on style as expressed in political speech. Style may, of course, manifest in other forms of legislative behaviour. For instance, prior work from other legislatures suggests that, relative to men, women are more collaborative (Holman and Mahoney, 2018); are more likely to initiate and sponsor legislation (Volden, Wiseman and Wittmer, 2013); and are more responsive to their constituents (Anzia and Berry, 2011). While our findings from the UK clearly do not speak to these other types of behaviour in other settings, many of these studies nevertheless document gender differences at a fixed point. Our results suggest that it is important for work in this field to pay more attention to how gendered dynamics evolve over time.

From a methodological perspective, we argued that existing approaches to detect-

ing style suffer from a well-known problem for off-the-shelf dictionaries: the words that demonstrate a given concept in one context may be poorly suited to detecting the use of that concept in another context. We circumvented this problem by using a word-embedding model which helps capture how different political styles manifest in the specific setting of parliamentary debate. Results from our validation show that this approach significantly outperforms existing methods, a finding that we think justifies adoption in other settings. Our strategy is likely to be useful whenever researchers are interested in measuring a latent concept from a large corpus of texts, but where the domain of interest differs from the domain in which existing dictionaries were developed. This describes a large fraction of applications of dictionary methods in political science, and so our approach has the potential to be applied widely and profitably elsewhere.

We argued that the stylistic convergence of men and women in parliament reflects the declining relevance of gender stereotypes in UK politics over time. While we were able to rule out two important alternative explanations, the nature of our data prevents us from providing strong evidence regarding the mechanisms that led to these changes. For instance, voters are less likely to endorse traditional gender stereotypes now than in the past, but we lack data to assess whether there has been a concomitant decline in the sanctions that voters apply for gender-role-inconsistent behaviour. Future work on these issues should focus on collecting over-time survey data on voters' attitudes towards non-stereotypical behaviour by politicians, and on estimating *local* opinion on these issues which might then be linked to the behaviour of specific MPs in parliament.

Similarly, we suggested that one important development relevant to the power of gender stereotypes is the increased prominence of women in parliament in recent years. However, the number of female MPs has increased monotonically over time in the UK, and so it is hard to distinguish the effect of this type of descriptive representation from other trends. Future work might use a design in the spirit of Karpowitz and Mendelberg

(2014), who randomly assign participants in political discussion to groups with varying gender compositions. Do gender-stereotypical communication styles decline when more women are present in a group? We hope that future work can help to shed more light on the mechanisms that underpin the trends that we document here.

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Appendix – The Declining Importance of Gender Stereotypes for Politicians' Style in the UK *

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*__This version__: September 24, 2020.

Word-embedding-based dictionaries

Our word-embedding-based measurement strategy consists of a number of steps, which we describe in more detail in this section.

First, for each style we define a "seed" dictionary that represents our concept of interest. We use the following sources to construct our seed dictionaries:

1. **Affect** – Linguistic Inquiry and Word Count 2015 (Affect) (Pennebaker et al., 2015)
2. **Fact** – Linguistic Inquiry and Word Count 2015 (Number and Quantitative) (Pennebaker et al., 2015) and all occurrences of any numeric figures
3. **Positive Emotion** – Regressive Imagery Dictionary (Emotions: Positive Affect) (Martindale, 1990)
4. **Negative Emotion** – Regressive Imagery Dictionary (Emotions: Anxiety and Sadness) (Martindale, 1990)
5. **Aggression** – A bespoke dictionary of words
6. **Human Narrative** – A bespoke dictionary of words and the 200 most common names of children born between 1970 and 2019

The final two seed dictionaries – which relate to aggression and human narrative – are our original constructions. These dictionaries were constructed by reading and watching debates from the House of Commons that are known to feature either aggression (for instance, Prime Minister's Questions) or examples of human narrative (for instance, debates on mental health or social policy issues), and selecting words and phrases that we thought were likely to capture these concepts in a broader set of parliamentary debates. We report the full lists of words that feature in these new seed dictionaries in figures S1 and S2

Second, a key component of our approach to measuring style are a set of word-embeddings, which we estimate from the full corpus of parliamentary speeches. Word-

Figure S1: “Aggression” seed dictionary

irritated ; stupid ; stubborn ; accusation ; acuse ; accusations; accusing ; anger ; angered ; annoyance ; annoyed ; attack ; insult ; insulting ; insulted ; betray; betrayed ; blame ; blamed ; blaming ; bitter; bitterly ; bitterness ; complain; complaining; confront ; confrontation; fibber; fabricator ; phony ; fibber ; sham ; deceived ; deceive ; disgrace; villain; good-for-nothing; hypocrite ; deception; steal ; needlessly; needless; criticise ; criticised ; criticising ; blackened ; fiddled; fiddle; problematic ; lawbreakers ; offenders; offend; unacceptbale ; leech; phoney ; appalling ; incapable ; farcical ; absurd ; ludicrous; nonsense ; laughable ; nonsensical ; ridiculous; outraged ; hysterial ; adversarial ; aggressive ; shady ; stereotyping; unhelpful ; unnatural ; assaulted ; assault ; assaulting ; half-truths ; petty; humiliate ; humiliating ; confrontational; hate ; hatred ; furious ; hostile ; hostility ; nasty; obnoxious ; sleeze; sleezy ; inadequacy; faithless; neglectful ; neglect; neglected; wrong ; failure ; failures ; failed ; fail ; scapegoat ; cruel; cruelty ; demonise ; demonised ; tactic ; trick; trickery ; deceit ; dishonest ; deception; devious; deviouness; shenanigans ; fraudulence ; fraudulent ; fraud; swindling; archaic ; sly; slyness; silly; silliness ; scandal; scandalous ; slander ; slanderous ; libellous ; disreputable ; dishonourable ; shameful; atrocious ; gimmick ; immoral; ridicule; antagonistic ; antagonise ; ill-mannered; spiteful ; spite ; vindictive ; prejudice ; prejudices ; disregard ; arrogant ; arrogance ; embarrasment ; embarrass; embarrasing ; distasteful ; provoke; provoked ; petulant ; ignorance ; stupidity ; idiot ; idiotic ; annoying; dodgy ; untrue ; penny-pinching ; attacking ; ironic ; irony ; outrageous; hackery; crass; backchat; rude ; ill-judged ; ragbag; mess; hash ; fiasco; shambles ; shambolic ; farce; botch; botched ; blunder ; mischievous; mischief ; undermine ; straightjacket ; groan; abuse; chaos; chaotic ; dull; predictable ; negligent; grotesque; scapegoats; hypocrisy; bogus; counterproductive; betrayal; patronise ; patronising; reprehensible; fool; foolish; abysmal ; disgraceful; woeful; inferior ; sneaky ; scaremongering; scaremonger; coward; cowardly; ignorant; intolerant; unacceptbale ; condemn; short-sighted; ashamed; falsehood; blackmail; clownery; debased; debase; hypocracy; mislead; misleading; smokescreen; subterfuge; horrendous; despicable; deplorable

Figure S2: “Human narrative” seed dictionary

example; constituent; person; someone; instance; surgery; case; told; illustrate; anecdote; experience; people; individual; cases; man; woman; mother; father; son; daughter; uncle; aunt; cousin; wife; husband; parent; child; say; said; support; discuss; speak; community; local; area; family; issues; remember; recall; married; resolve; authorities ; help; imagine; envisage; lives; sometimes; concerned; heard; circumstance; anyone; nobody; citizens; relationship; girl; boy; believe; listen; problem; inspire; many; comment; authority; conversation; worked; tell; thought; life; home; referred; situation; happened; everyone; concern; recognise; advice; advise; everyday; personal; letter; involve; nephew; niece; learn; local area; my constituents; previous job; tell me; told me; first hand; speaking as; own experience; for example; I recognise; I remember; help people; many years; see me; spoke with; their; them; talk; constituency ; constituents ; mum; dad; rhetoric; mr ; mrs ; know ; wrote ; write; ask ; call; dr; doctor; society; ordinary ; together ; dear; honest; visit; everybody; feel; view; public ; employer ; reflect; born; expect; anybody; responsibility ; youngster; heartbreaking; young; hopeless ; desperate; picture; chat; electorate; provide for; foster; colleague; represent ; neighbourhood; locality ; sympathy ; condolence; grief; bereavement ; trust; serve; communicate; testimony; motherhood; fatherhood; sensitive; remark; couple; brave; lifelong; proud; pride; facilities; quote; real; meet; met; childhood; reminisce ; nostalgia; recollect; hometown; lifetime; email; neighbour; partner; children; teenager; youth; contact; tale; scenario; bred; hard-working; year-old; friend; parent; parents; came; knew; recently; lady; gentleman; families

embedding models, which are of increasing use in political science (Spirling and Rodriguez, 2019), seek to describe any word in a corpus as a dense, real-valued vector of numbers. The construction of the word-embedding vectors, regardless of the specific algorithm used to estimate them, relies centrally on the distributional hypothesis: the idea that words which are used in similar contexts will have similar meanings. Here, a context refers to a window of words around a target word, and the embedding model allows us to *learn* the semantic meaning of each word directly from the use of the word in the corpus.

The main output of embedding models are the word-embeddings themselves. These are vectors that correspond to each unique word in the corpus. The dimensions of the embedding vectors capture different semantic "meanings" that can be used to provide structure to vocabulary. Crucially for our purposes, given this representation, the distances *between* word-vectors have been shown to effectively capture important semantic similarities between different words (Mikolov et al., 2013). We use this property to define the set of words that, *in the context of parliamentary debate*, are used in a semantically similar fashion to the seed words.

We follow the estimation procedure outlined in Pennington, Socher and Manning (2014) and estimate a word embedding, W, of length $J = 150$ for each unique word in our corpus. We use a small "context" window size of 3 words either side of the target word to estimate our embeddings. This is consistent with our aim of capturing semantic (rather than topical) relations between words (Spirling and Rodriguez, 2019, 7). We exclude all words that occur very rarely (fewer than 90 times overall), and all words that occur very frequently (in more than 90% of documents). We remove all stop-words, punctuation, and a list of parliamentary address terms such as "Honourable Friend" or "Home Secretary". We collect the embeddings in a matrix, θ, which we use to calculate the mean word-embedding vector for each of our seed dictionaries. The average word-embedding

of the seed words represents the "location" of the dictionary in the vector-space defined by the embedding model, and allows us to calculate the relative semantic similarity of different words to the dictionary.

Third, we calculate the similarity between *every* word in the corpus and the mean dictionary word-vector using the cosine-similarity metric. Words closely related to the average semantic meaning of the seed words will have a high similarity score, and words that are less closely related will have a low similarity score. We then follow Zamani and Croft (2016) and apply the sigmoid function to the similarity scores, which transforms all similarity scores to the [0,1] interval and shrinks the scores of all but the most similar words to very close to zero. Where x_w^s is the cosine similarity between the word-embedding for word w and the mean word-embedding of the seed dictionary for style s, the sigmoid transformation is given by:

$$Sim_w^s = \frac{1}{1 + e^{-a(x_w^s - c)}} \tag{S1}$$

Here, a and c are free parameters which we set to be equal to 40 and .35, respectively, based on the results in Zamani and Croft (2016, 3). Sim_w^s gives our final score for each word for each style. Words closely related to the average semantic meaning of the seed words for a given dictionary will have a high Sim_w^s, and words that are less closely related will have a low Sim_w^s.

Finally, we use the word-level scores, Sim_w^s, to score each *sentence* in the corpus. As described in the main body of the paper, the score for a given sentence on a given dimension is:

$$Score_i^s = \frac{\sum_w^W Sim_w^s N_{wi}}{\sum_w^W N_{wi}} \tag{S2}$$

where Sim_w^s is the similarity score defined above, and N_{wi} is the (weighted) number of

times that word w appears in sentence i, where the weights are term-frequency inverse-document-frequency weights.[1] $Score_i^s$ represents the fraction of words in sentence i that are relevant to dictionary s. When words with high scores for a given style appear frequently in a given sentence, the sentence will be scored as highly relevant to the style. The score for each *document* is then the weighted average of the relevant sentence level scores, where the weights are equal to the number of words in each sentence.

[1]TF-IDF weighting is used to down-weight very common words, and up-weight relatively rare words.

Validation tests

As with all QTA approaches, careful validation of our measures is essential (Grimmer and Stewart, 2013), and we provide two face validation checks in this section, as well as more detail on the human validation task that we describe in the main text.

Word- and sentence-validity

First, in table S1, we examine the words that are associated with large Sim_w^s values for each of our styles. In particular, the table shows the top 30 words associated with each concept according to our word-embedding measure (*Top*), the words that are high-scoring based on the word-embedding measure but which do not feature in the seed dictionaries (*Added*), and the words that are low-scoring on the word-embedding measure but which did feature in the seed dictionaries (*Removed*). The *Added* words are particularly important, as they represent words that are used in a similar context to the words in our seed dictionary in the parliamentary setting but which would be missed by traditional dictionary based approaches.

The tables reveal that high-weight words (*Top*) generally correspond very closely to the style dimensions to which they relate. For instance, the top-loading words in the "Positive Emotion" dimension include "joy", "delight", "eager", and "excitement". Similarly, in the "Aggression" dimension, top words include "disgraceful", "shameful", "outrageous", and "scaremongering". It is also encouraging that the top words in the "Fact" dimension are mostly numeric quantifiers, and the top "Human Narrative" words include "constituent", "told", "wrote", "said", and several words that indicate specific individuals ("son", "father", "wife").

In addition, many words that are not included in the original seed dictionaries are nevertheless given high weights via the word-embedding approach (*Added*). For exam-

ple, the words “shocking”, “incompetence”, “pathetic”, and “deplore” do not appear in the “Aggression” seed dictionary, but nevertheless receive high weights for that style. That these words are consistent with intuitive notions of these broad stylistic categories, although not in the original dictionaries, highlights the fact that the word-embedding approach is successfully finding words that are semantically closely related to our key concepts of interest.

Similarly, the table also shows that some words included in the original seed dictionaries which are not semantically similar to the relevant concepts in the context of parliamentary debate are given low weights by the word-embedding approach (*Removed*). For example, that “terrorist” is removed from the “Negative Emotion” dictionary is encouraging, as within a parliamentary context the use of the word “terrorist” is likely to be from a reference to matters of policy rather than to an expression of emotion.

Overall, the words in table S1 suggest that our word-embedding model is a) accurately associating sensible words with our stylistic concepts; and b) capturing language use that is representative of a given style, even when those words are not included in our seed dictionaries, and so would be missed by traditional dictionary approaches.

Affect			Positive Emotion		
Top	*Added*	*Removed*	*Top*	*Added*	*Removed*
feel	feel	award-winning	joy	eager	reliefs
afraid	really	securities	delight	pity	gladstone
really	sometimes	approvals	enjoyable	anticipation	satisfied
sometimes	frankly	destroyers	eager	love	relieve
fear	always	engagements	excitement	sadness	relief
frankly	undoubtedly	dwellings	enjoying	liked	relieving
always	think	admiral	celebration	amazed	satisfactorily
undoubtedly	nevertheless	festival	amused	horrified	satisfy
think	genuinely	super	relieved	appreciated	grind
nevertheless	often	approving	happy	birthday	gay
genuinely	feeling	shakespeare	cheer	wonderful	satisfactory
often	felt	approved	celebrate	look_forward	satisfies
feeling	seem	championships	delighted	admire	entertainment
felt	believe	treasurer	enjoyed	hear	grinding
seem	deeply	championship	pity	churlish	satisfaction
believe	indeed	pesticides	fun	praise	amusement
certainly	however	challenger	anticipation	sincerely	laughs
upset	something	approve	entertaining	informative	gladly
worry	say	energy-intensive	love	doubtless	laughed
deeply	perhaps	relaxing	enjoy	coincidence	cheers
indeed	probably	risk-based	sadness	saddened	laughing
however	find	harmonise	liked	fascinated	satisfying
something	somehow	resignations	amazed	lovely	cheered
say	sense	flexibilities	horrified	shocked	entertain
regret	quite	festivals	excited	fascinating	enjoyment
perhaps	nothing	glamorgan	appreciated	compliment	enthusiastically
probably	necessarily	laughs	celebrating	greeted	laughable
hurt	people	approves	amusing	always	joke
find	thought	awarded	birthday	astonished	happiness
somehow	thing	exhaustive	wonderful	pleased	rejoice

Negative Emotion			Aggression		
Top	*Added*	*Removed*	*Top*	*Added*	*Removed*
upset	upset	painstaking	disgraceful	cynical	inferior
suffering	terrible	painting	shameful	utterly	offenders
terrible	hurt	alarms	outrageous	embarrassing	assaulted
distressing	deeply	paint	scaremongering	frankly	annoyance
hurt	unfortunate	avoidance	cynical	misguided	fiddle
distress	angry	miserably	dishonest	pathetic	fiddled
unhappy	feeling	paints	utterly	incompetence	assault
worry	felt	terrific	embarrassing	irresponsible	furious
deeply	compounded	cowardly	frankly	bizarre	fail
dreadful	caused	terrorists	scandalous	dreadful	steal
frightening	horrendous	grievance	absurd	reckless	fool
unfortunate	shocked	terrorism	ridiculous	complacency	predictable
angry	horrible	lone	ludicrous	incompetent	deceived
anxiety	frustrating	hopelessly	misguided	illogical	problematic
suffer	appalling	disappointingly	deplorable	wicked	offend
despair	anger	alarmingly	pathetic	disingenuous	needlessly
frightened	frustration	grievances	incompetence	downright	neglected
suffered	appalled	terrorist	irresponsible	shocking	embarrass
worried	frustrated	discouraged	bizarre	complacent	fiasco
sad	experiencing	shy	appalling	deliberate	mischief
feeling	disturbed	painted	nonsense	deplore	deceive
felt	sometimes	timid	dreadful	callous	cruelty
tragic	unpleasant	avoids	reckless	unjust	bitterness
fear	embarrassing	pitiful	complacency	naive	provoked
horror	feel	alarmist	arrogant	hypocritical	dodgy
compounded	terribly	discouraging	ashamed	unfortunate	confrontational
caused	shocking	sufferers	hypocrisy	horrible	adversarial
horrendous	understandably	discourage	incompetent	plainly	failed
anguish	disturbing	avoiding	despicable	utter	archaic
shocked	uncomfortable	lamentable	illogical	folly	bogus

Fact			Human Narrative		
Top	*Added*	*Removed*	*Top*	*Added*	*Removed*
half	nearly	sixthly	told	like	poppy
nearly	years	sevenoaks	constituent	whose	bred
five	year	doubly	know	indeed	hopeless
four	whereas	infinitely	wrote	called	amber
ooo	per	ooost-century	said	says	recollect
six	£	oooth-century	constituents	think	florence
seven	months	scarce	mr	saying	georgia
eight	cent	double-dip	like	went	alice
three	just	infinite	father	just	anecdote
years	past	bunch	whose	also	aunt
year	days	groupings	son	others	spencer
quarter	ago	equalities	indeed	see	eve
whereas	almost	seven-day	met	former	albert
nine	compared	equalise	remember	perhaps	rhetoric
two	thirds	fifthly	came	week	skye
million	weeks	grouped	called	ooo-year-old	kate
approximately	next	group's	tell	obviously	chat
per	equivalent	samples	wife	wanted	ashton
least	roughly	ooo-page	david	described	mohammed
£	now	ooog	say	lord	tale
months	moreover	sixth-form	dr	mine	inspire
cent	time	oood	says	one	roman
just	figure	ooo-to-ooo	woman	friends	everyday
billion	hours	quarterly	think	knows	scenario
past	week	grouping	man	yesterday	locality
average	yet	triple	saying	unfortunately	nicola
days	spend	oob	went	understand	bereavement
ago	furthermore	equalisation	mother	looked	sensitive
third	last	six-year	friend	aware	illustrate
almost	number	six-week	just	ago	childhood

Table S1: Word-level validation

Tables S2 and S3 assess the face validity of our approach by showing the 10 highest scoring *sentences* for each style, according to the $Score_i^s$ measure described in equation S2. For all styles, the sentences clearly reflect the conceptual definitions outlined above. For instance, the "fact" category is dominated by statements using numerical language, and the "human narrative" category has many examples of MPs referring to the experiences of specific individuals. This again suggests that our measurement strategy plausibly captures our stylistic dimensions of interest.

Table S2: Top sentences for Affect, Positive Emotion, and Human Narrative

Affect	Positive Emotion	Human Narrative
In others, everyone seems a little depressed - perhaps not greatly upset but a little depressed none the less.	As always, it is an enormous pleasure to follow the hon Member for Bootle , whose speeches are always entertaining and occasionally informative.	Moreover, what happens when an elderly brother and sister live together, or an elderly mother lives with her elderly son?
Others eventually got jobs, although usually far less rewarding, far less secure and far less well paid.	It is always a pleasure to listen to Members' maiden speeches, and I enjoyed his as well.	Last week a friend of mine who works with elderly residents in Ogmore visited four elderly residents in one day.
Some of us believe that the legislation is profoundly unacceptable, profoundly wrong and profoundly damaging to our country.	I am always excited and in a state of eager anticipation to hear what the right hon Gentleman has to say on everything.	Anyone whose wife or partner had a child 20 years ago will remember that the woman spent a week to two weeks in hospital.
People understandably already feel fraught and upset - they are in a situation that they never anticipated, and feel vulnerable and sometimes deeply hurt and angry.	I also congratulate my hon Friend the Member for Blackpool, North on his most amusing, entertaining and sincere maiden speech.	On Monday this week, another south Birmingham MP and I met South Birmingham primary care trust to talk about the situation in south Birmingham.
Many mentally ill people face sad and painful lives with great courage - more courage than the rest of us may have.	I begin this afternoon by wishing the Secretary of State a very happy birthday - I sincerely hope that it improves from here on.	I have also discovered that a person called Mr Richard Shires subsequently became a paid constable in West Yorkshire police and continues to serve to this day.
I am naturally sorry to hear of my noble Friend Lord Archer' s unhappy experience, although clearly there is rather less sympathy among Opposition Members.	I join hon Members across the House in wishing a happy Pride to all those celebrating London Pride this weekend.	However his father David suffered a stroke 13 years ago since when his mother Sarah has had to care for both son and husband.
But neither can anyone underestimate the anger and sadness among people that things should ever have been allowed to get into this position.	It was wonderful to hear the shadow Chancellor - it is always wonderful to hear the shadow Chancellor in his marvellous speeches - explaining how cross-party he was.	I speak as someone whose father served in the Metropolitan police for 25 years and whose younger brother is a serving Metropolitan police officer.
That would give an awful lot of confidence to an awful lot of pensioners who feel hard done by.	Today's debate has been extremely lively, interesting and, at times, amusing and much good wit and humour have made it a delight.	My constituent's father goes on: Two years ago his wife walked out with another man taking the two children.
Such serious problems have left many facing uncertainty, which can cause severe stress to people who already face incredibly challenging circumstances.	The Minister is nodding; I am grateful for that, and I look forward with anticipation to his reply.	American civilians took leave once every six months; British diplomats took leave every six weeks, for two weeks.
Prisoner releases continue - causing, I must say, understandable anguish among many victims' families, but going ahead none the less.	For that reason, I wish him a very happy birthday and hope that his family enjoy their celebrations.	Let us imagine that the parent of a 14-year-old girl becomes aware that she is seeing a much older man.

Table S3: Top sentences for Aggression, Fact, and Negative Emotion

Aggression	Fact	Negative Emotion
I found the attitude of the Conservatives' motion not only hypocritical and incoherent, but profoundly cynical and dishonest.	The maximum figure for those costs was $91 billion, although the real extra costs amounted to $26 billion.	People understandably already feel fraught and upset - they are in a situation that they never anticipated, and feel vulnerable and sometimes deeply hurt and angry.
That statement is as barbaric as it is downright stupid; it is nothing more than an ignorant, cruel and deliberate misconception to hide behind.	None the less, social security on average now costs every working person nearly £15 every working day.	However, the indignity, discomfort and inconvenience caused to Brian during this episode understandably left him feeling demoralised and, in his words, depressed.
It is grossly irresponsible and, I am afraid, profoundly and disturbingly misleading, and even ignorant, to go around doing that.	Recent figures show the current account deficit running at the much lower level of £0.5 billion per month.	This is deeply worrying for families living in those blocks, and is causing huge anxiety, fear and insecurity.
Of course the situation in Zimbabwe is disgraceful and we condemn utterly the barbaric attacks on farmers, which are totally unacceptable.	The annual average income of the poorest fifth of households has risen by £700 in real terms since 2007-08.	In addition to suffering horrendous physical injuries, enormous physical stress and emotional trauma, they had enormous financial stress.
There is something horrible, vindictive and cowardly about the Government's intolerant and ignorant attack on a small minority".	We now spend nearly £11 billion extra each year on pensioners, and almost half that additional spending goes to the poorest third.	It can cause misery and pain for individuals and their families through serious disease or, worse, death.
They should not be all about blaming people, because blaming individuals for errors and mistakes is unhelpful and counter-productive.	Women in Wales earn three quarters of what men earn - even though men in Wales are among the lowest paid in Britain.	To the families we say: we are deeply sorry for your loss and deeply sorry for the pain you have suffered.
Some of us believe that the legislation is profoundly unacceptable, profoundly wrong and profoundly damaging to our country.	A quarter of people who become unemployed leave unemployment within a month, half within three months and about two thirds within six months.	Their anger is the anger of pain, the anger of discrimination, and the anger of lack of understanding, as well as the anger of frustration.
Worse even than the failure publicly to criticise and condemn has been the United Kingdom Government's tendency almost to excuse.	The five Conservative speakers took three hours, five minutes; the six Labour speakers took one hour forty-five minutes.	The grief and stress caused by the death of a loved one are prolonged and added to by financial worry and hardship.
If that happens because of an arrogant and incompetent subordinate should not that arrogant and incompetent subordinate be fired?	The growth rate figures are substantially different from the growth rate figures produced in the Budget just four months ago.	Lack of sleep, nervous distress, illness and severe depression have all been experienced by people who have been subjected to inconsiderate and noisy neighbours.
As one of the four Ministers, I find that accusation deeply offensive, deeply insulting and entirely without foundation.	Lower than expected unemployment is already saving around £10 billion over the next five years on benefit spending alone, compared with Budget plans.	If people feel isolated, depressed, lonely, jobless and skill-less, they will feel worse in hospital.

Human validation task

In the paper, we provide results from a human validation task which assesses whether our text-based measures of style mirror human judgments of the same concepts. We wrote a web app which presented our coders with pairs of sentences (sampled from all sentences in our corpus). Coders were asked to complete two tasks. First, a style-*comparison* task required them to select which of the two sentences was more typical of a particular style. Second, a style-*intensity* task required them to rate the degree to which each sentence was representative of the selected style on a 5 point scale.

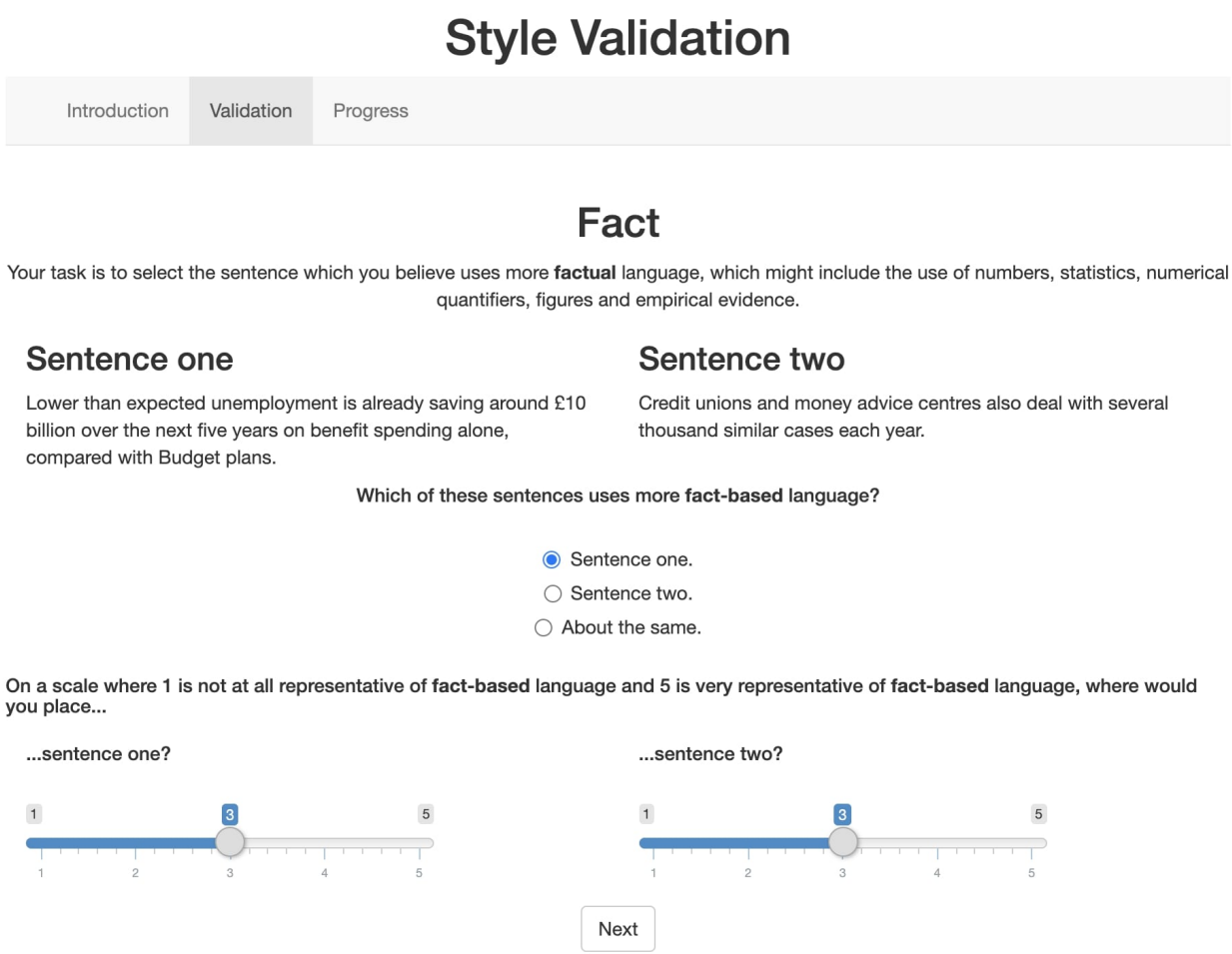

Figure S3: Human validation task prompt

Figure S3 gives an example of the prompt seen by our coders. In addition to the sen-

tences themselves, we presented coders with minimal definitions of the speech-styles of interest in order to ensure that the human coding related to the style dimensions identified in the literature review.

Each coder completed 70 comparisons per style, on average, meaning that we have on average 140 individual sentence-ratings per style. We use the distribution of responses to these tasks, and compare them to the distribution of text-based style measures described in the main body of the paper for the same sentences as seen by the coders.

Figure S4 presents the results from the pairwise-comparison task. On the x-axis we present the standardized difference in the word-embedding based measures for the sentence pairs, where positive numbers indicate that the style is more prominent in sentence one, and negative numbers indicate that the style is more prominent in sentence two. The y-axis shows the equivalent decisions from the human coding task, where "Sentence one" indicates that the human-coder judged the style to be more prominent in the first sentence, and "Sentence two" indicates that the human-coder judged the style type to be more prominent in the second sentence, and "About the same" indicates that the human-coder judged the use of the style to be roughly equivalent in both sentences. Black lines are smoothed loess curves.

Figure S5 present the results from the intensity validation task. On the x-axis are the word-embedding measures for each sentence, where higher numbers indicate the sentence is more representative of the style type. The y-axis shows the equivalent decisions from the human coding task, where a score of 1 indicates the coder judged a sentence to be not at all representative, and a score of 5 indicates the coder judged a sentence to be extremely representative. Again, black lines are smoothed loess curves.

The plots reinforce the central message in the validation section of our paper: the scores derived from our word-embedding based approach are strongly positively correlated with scores assigned by human readers of the same sentences. Moreover, as

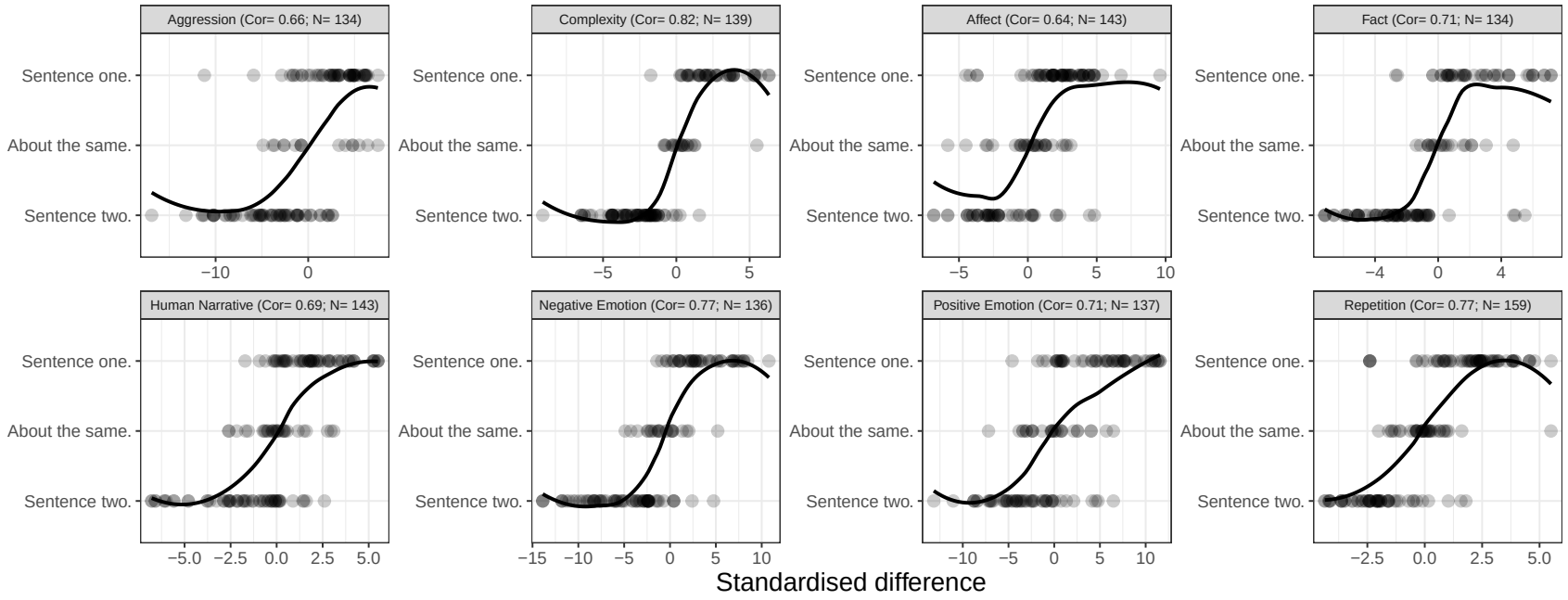

Figure S4: "Comparison" validation task

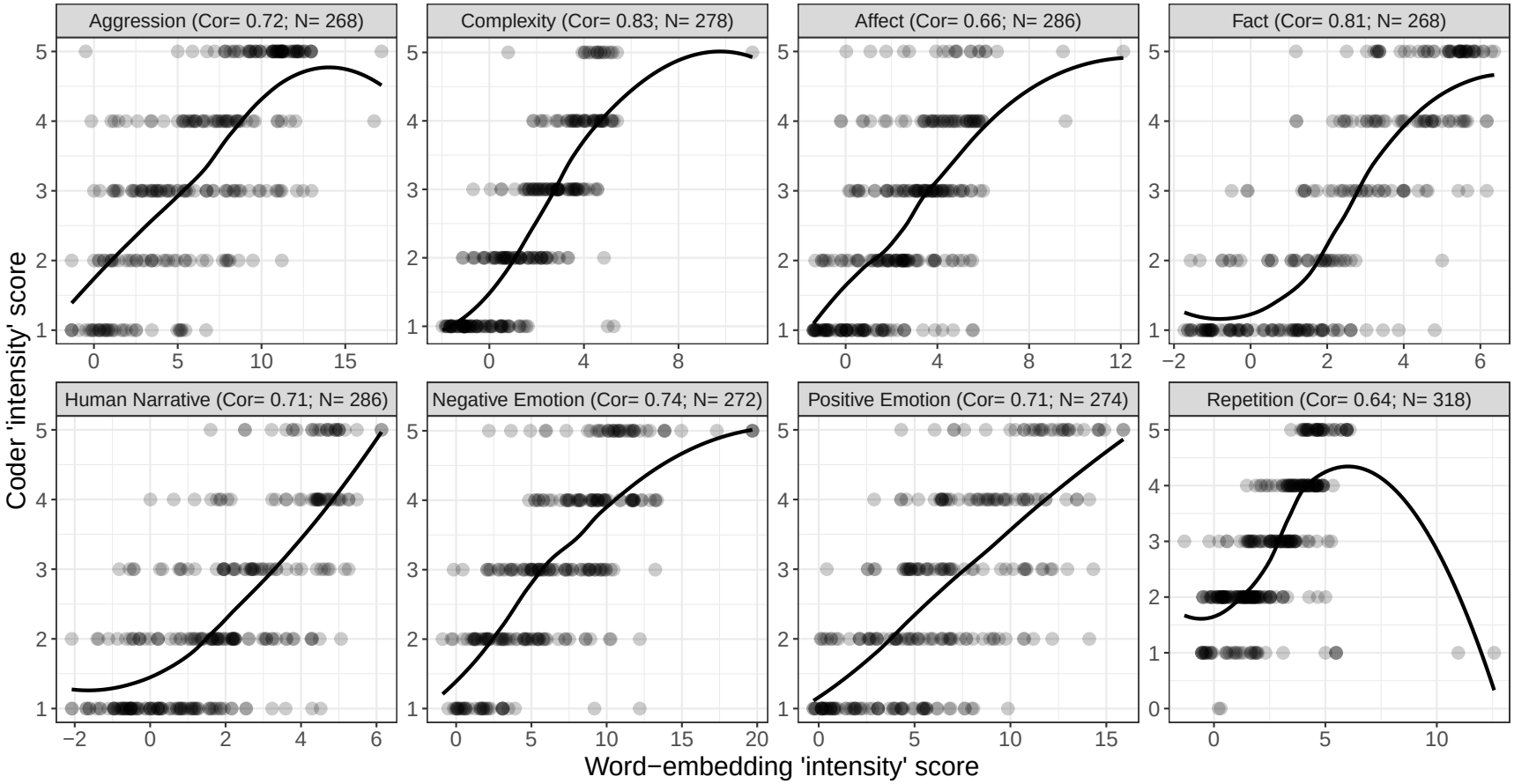

Figure S5: "Intensity" validation task

we show in the main body of the paper, these correlations are higher than they are for comparisons between human judgments and standard dictionary-based measurement approaches.

Controlling for individual-level covariates

In this section we show results of the alternative specification for the dynamic hierarchical model described in the paper in which we expand the model at the second level by including a vector of individual-level covariates, $X^k_{j,t}$:

$$\alpha_{j,t} \sim N(\mu_{0,t} + \mu_{1,t} Female_j + \sum_{k=1}^{k} \lambda_k X^k_{j,t}, \sigma_\alpha) \tag{S3}$$

where $X^k_{j,t}$ includes dummy variables for party; whether an MP holds a government or opposition frontbench position; whether an MP was a committee chair in the session; whether an MP was a member of one of the governing or opposition parties; and continuous indicators for MP age and the number of years served in parliament. We present the results for our main quantities of interest ($\mu_{1,t}$) estimated from this model in figure S6.

The figure shows that, in general, we recover very similar patterns of gender differences in style use over time when controlling for individual-level covariates. In particular, for human narrative, affect, positive emotion, negative emotion, fact, and repetition the trajectories of the gender differences over time are almost indistinguishable from those presented in the main body of the paper. The largest difference are for complexity and aggression, where the pattern of convergence between men and women is somewhat attenuated in the estimates from the alternative specification. We still see that women become more aggressive relative to men over time, though this change is smaller in these estimates than in those presented in the paper. For complexity, it is clear that the large shift in the gender difference that we observe between 2008 and 2013 is confounded by some of the individual-level covariates, as the gender difference is largely constant (and indistinguishable from zero) for the entire time period once we

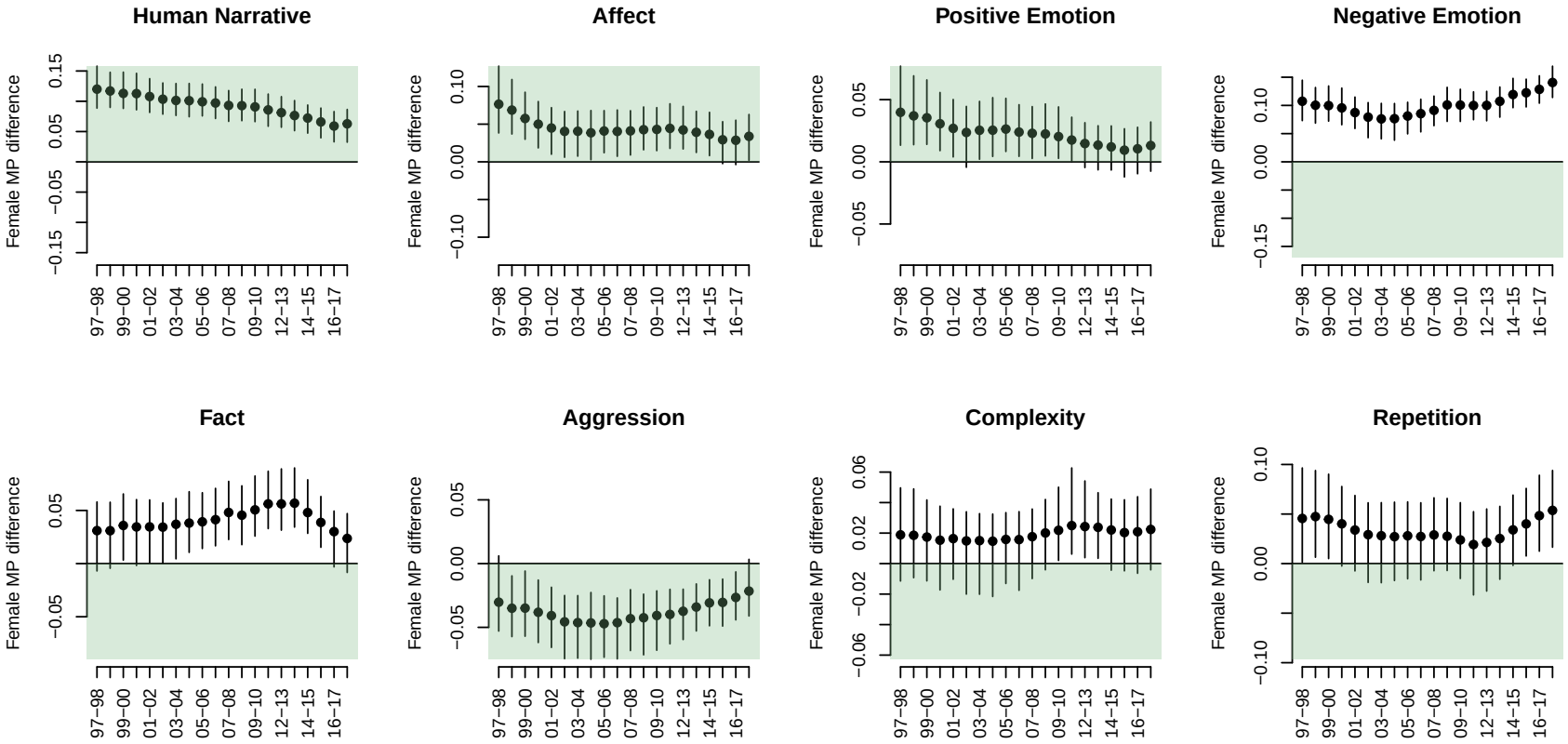

Figure S6: Gender differences in style over time controlling for individual-level confounders

control for these other factors. Nevertheless, overall, these results suggest that while other MP-level characteristics clearly account for variation in style use, our central finding – that the debating styles of male and female MPs have diverged from gender-based stereotypes over time – is not affected by these estimates.

Figure S7 presents the estimates for each of the individual-level covariates for each style. Although these are not our primary quantities of interest, there are several patterns that are of substantive interest. First, we find, consistent with other work (Proksch et al., 2019), that MPs from government parties use significantly less negative and more positive language than MPs from opposition parties. Government MPs are also less aggressive, and tend to rely more on human narrative and less on fact-based arguments than their opposition counterparts. Second, compared with backbench MPs, politicians in leadership positions are less likely to use human narrative, more likely to make fact-based arguments, use substantially less emotive language, and are more repetitious in their speeches. Finally, we see some evidence of partisan differences. Compared to Conservative Party MPs, Labour MPs use more human narrative, more factual language, and are somewhat less complex in their speeches. Liberal Democrat MPs, by contrast, make less use of human narrative, more use of fact, and are substantially less aggressive than their Conservative colleagues. Overall, it is clear that there are many factors that influence the political styles that MPs adopt and, while these are not directly relevant to the substantive questions in our study, we think that these finding may be profitably investigated in future work.

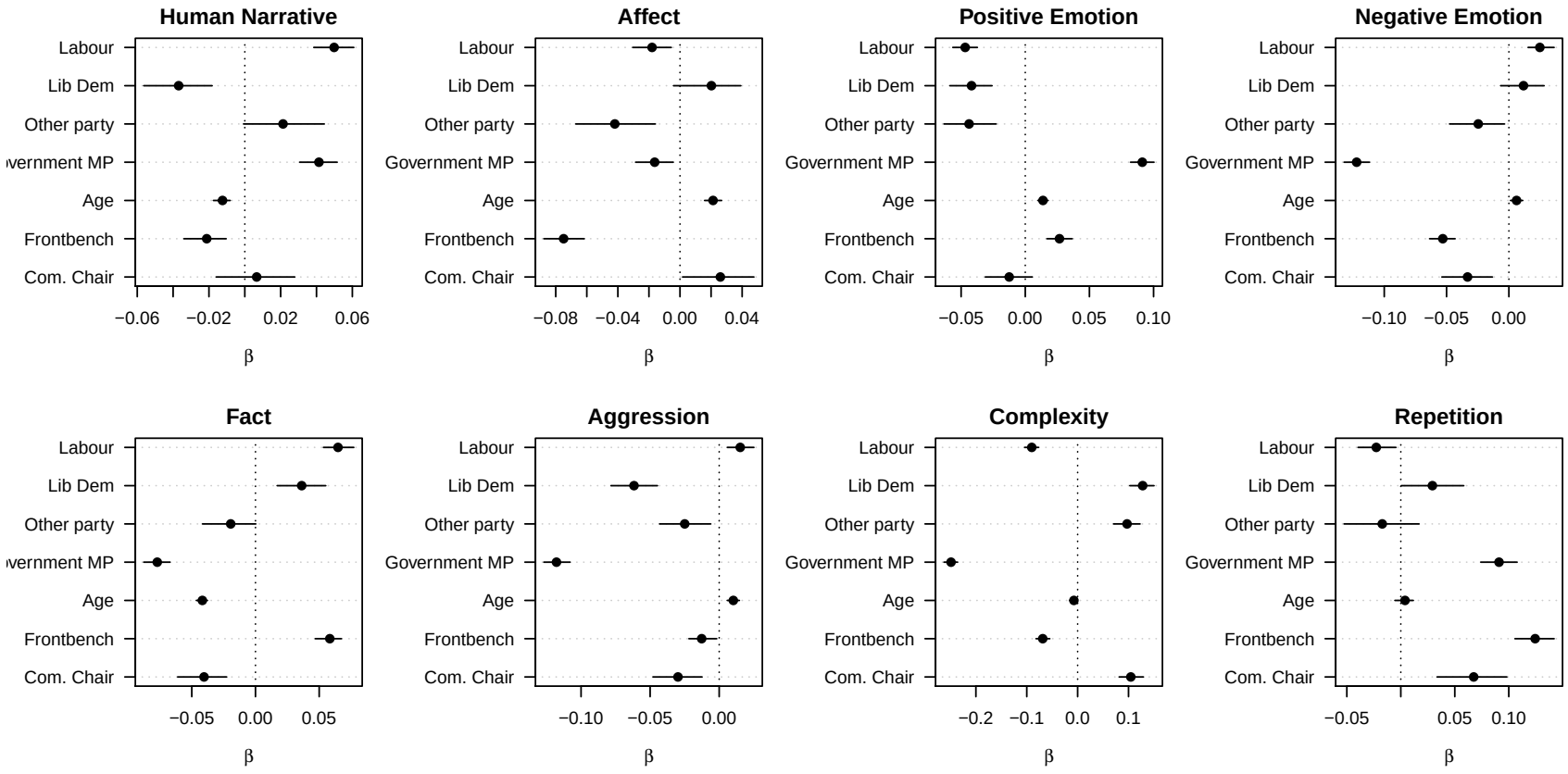

Figure S7: Individual-level covariate effects

Style use and debate-type

Our model accounts for aggregate differences in style use across debates via the δ_d random-effects described in equation 2. The inclusion of these parameters means that gender differences in style use cannot be attributed to men and women participating in systematically different types of debates, as the gender effects we estimate are based on within-debate variation in the style outcomes. However, it is possible that the magnitude of gender differences nevertheless varies across debates of different types. We investigate this possibility here. Specifically, we separate the debates in our data into common types that occur regularly in the UK House of Commons (for more detail, see Blumenau and Damiani, 2020):

1. **All**: all debates in our dataset.
2. **Ministerial Question Time**: the routine questioning of Ministers, occurs four times a week.
3. **Prime Minister's Question Time**: the Prime Minister answers questions from the Leader of the Opposition, opposition members and government backbenchers, occurs once a week.
4. **Procedural debates**: a compound category that includes debates that are not substantive in nature, but deal with matters of parliamentary procedure or scheduling. For example, Business of the House or Points of Order.
5. **Legislation**: debates on legislation, includes all stages of the process that occur in the Commons' chamber, such as second and third reading.
6. **Opposition Days and Backbench Business**: this includes business for debate that is placed on the parliamentary agenda by opposition members or backbenchers.
7. **Other**: all other forms of debate that are not captured by the above categories.

This categorisation captures important substantive differences between different types

of debates in the House of Commons, some of which have been shown to be predictive of MPs' style in previous work (Osnabrügge, Hobolt and Rodon, 2020).

We run a series of OLS models for each of our outcomes, where our main explanatory variable of interest is the gender of the MP, and where we also control for party, age, years in parliament, and whether the MP was a) a member of the cabinet, b) a member of the shadow cabinet membership, c) a government minister, d) a shadow minister, or e) a committee chair. For each outcome, we subset the data to only debates of a certain type, estimate the model, and record the coefficient on the gender variable at each iteration. Figure S8 shows, for each style, the gender differences in the seven different debate types.

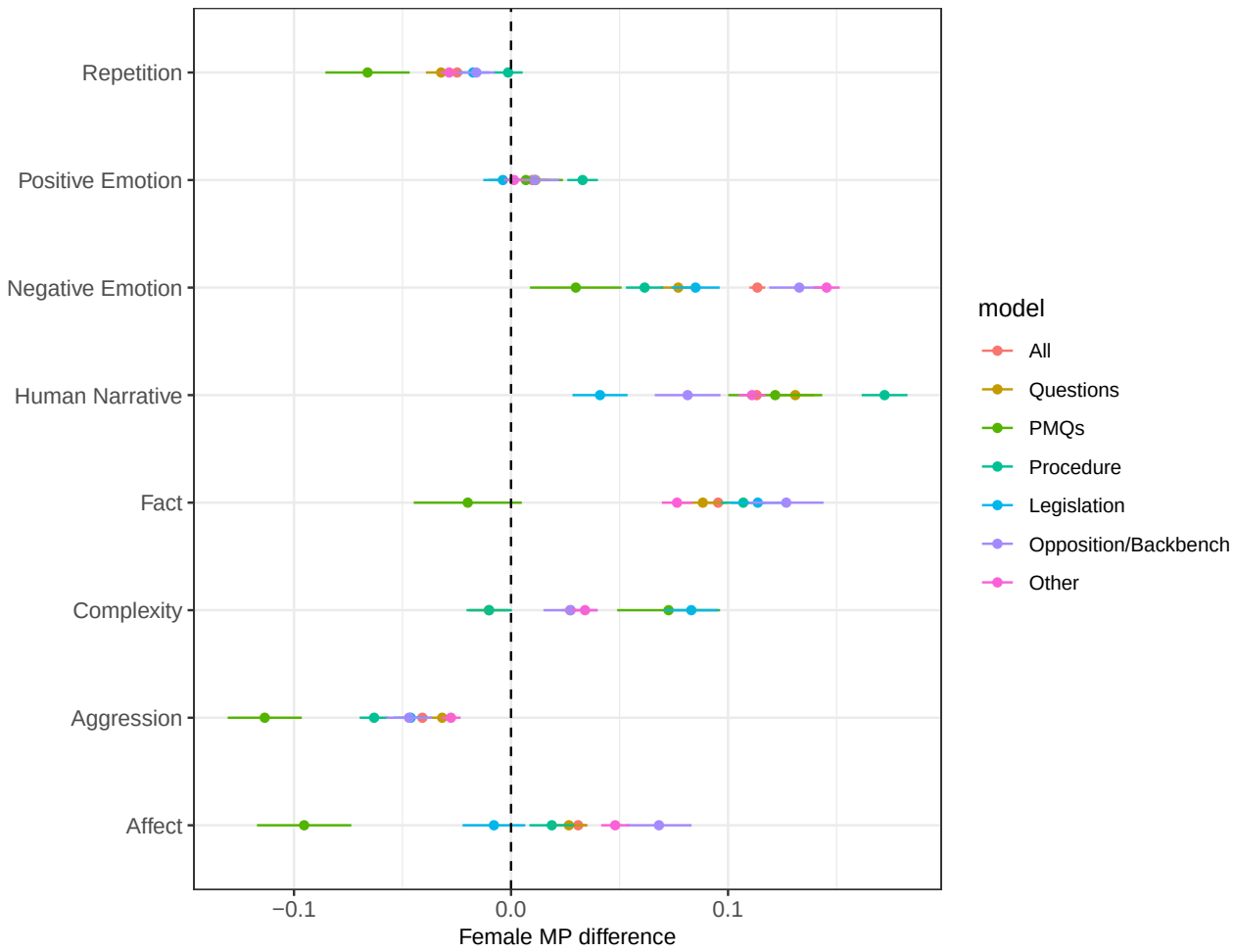

Figure S8: Debate type models

The analysis reveals that the magnitude of average gender differences are relatively constant across the debate types. In the debate types we identify, Prime Minister's Questions seems to be the only type of debate that significantly effects the gender coefficients. We see that, relative to the model which pools across all debates, the magnitude of gender differences is increased for repetition, aggression, and affect; decreased for negative emotion; and reduces gender differences in fact to statistically indistinguishable from zero. Overall however, while there is some variation in the magnitude of gender differences across debate types, these differences are for the most part very small.

In figure S9 we show additional descriptive information on the average level of each style in speeches used across the different debate types. The patterns in style use across debates generally conform with standard intuitions. For instance, the figure shows that both Question Time and Prime Ministers Questions (PMQ) debates are substantially less positive than debates on legislation, which is consistent with the idea that these settings are used by the opposition parties to interrogate - and often castigate - the government on issues of the day. Similarly, both PMQ debates and debates initiated by the Opposition parties in parliament are more aggressive than other debates, which again follows the intuition that these debates are mainly used as a vehicle for criticizing government policy. In general, these descriptive figures bolster the results from our validation exercises above, as they imply that our measures accurately capture expected differences in speech style across different types of parliamentary debate.

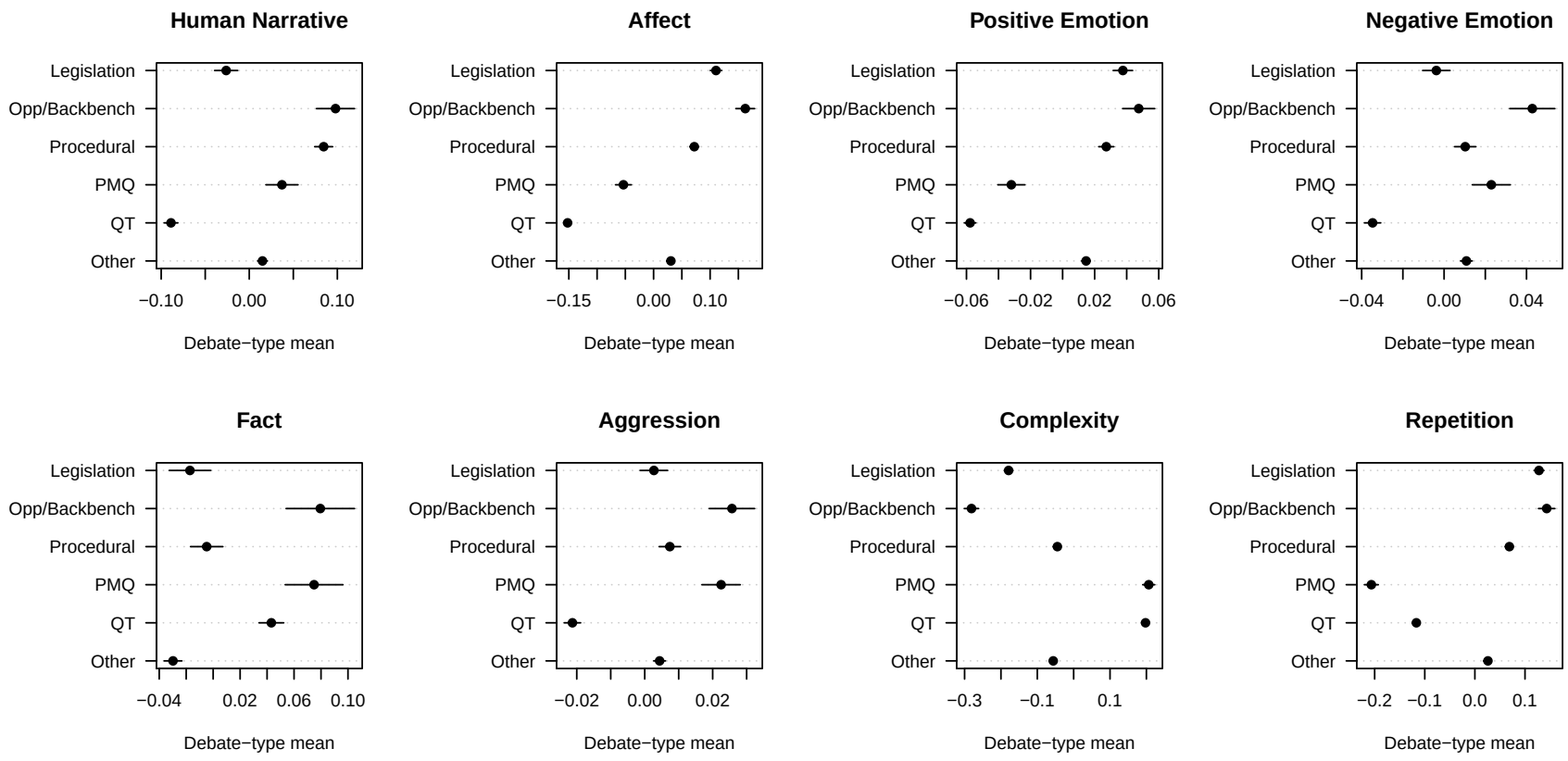

Figure S9: Style type average by debate type

Within-MP and replacement effects

Does gender explain less variation in aggregate style use over time because of a gradual convergence in styles of female and male MPs throughout their careers in parliament? Or do gender gaps decrease because the men and women entering parliament over time are systematically different from those leaving parliament? Which of these two explanations – which we refer to as "within-MP" and "replacement" effects – is responsible for the aggregate patterns we document in the main body of the paper? Our modelling approach allows us to decompose the evolving gender differences that we report in the section above into these two mechanisms of change.

Given the model described by equations 2 and 3 in the main body of the paper, we can decompose the shifting patterns of gendered style use into those changes that stem from within-MP change over time, and those that come from replacement. Our goal is to specify a decomposition of $\mu_{0,t} - \mu_{0,t-1}$, which is the change in average style use for men between parliamentary session t and session $t-1$ (we can then provide an equivalent approach for female MPs). We begin by distinguishing between three types of MP, which we label as "remainers", "joiners", and "leavers":

- J_m^R is the set of male MPs who appear in both session t and $t-1$ (Remainers)
- J_m^J is the set of men who appear in t but not in $t-1$ (Joiners)
- J_m^L is the set who appear in $t-1$ and not in t (Leavers)

We also will require the fraction of men who are "remainers" in t and $t-1$:

- π_t^R is the fraction of male MPs in t who also served in $t-1$
- π_{t-1}^R is the fraction of male MPs in $t-1$ who also served in t

Note that the proportion of male MPs who are "remainers" in t may be different from the proportion in $t-1$, because some male MPs who leave parliament in $t-1$ will be replaced by women in t (and vice versa).

Given these definitions, we can write the mean style use for men in each period as a function of the MP-period effects ($\alpha_{j,t}$):

$$\mu^m_{0,t-1} = \underbrace{\pi^R_{t-1}\frac{1}{|J^R_m|}\sum_{j\in J^R_m}\alpha_{j,t-1}}_{\text{Remaining MPs}} + \underbrace{(1-\pi^R_{t-1})\frac{1}{|J^L_m|}\sum_{j\in J^L_m}\alpha_{j,t-1}}_{\text{Leaving MPs}} \tag{S4}$$

$$\mu^m_{0,t} = \underbrace{\pi^R_{t}\frac{1}{|J^R_m|}\sum_{j\in J^R_m}\alpha_{j,t}}_{\text{Remaining MPs}} + \underbrace{(1-\pi^R_{t})\frac{1}{|J^J_m|}\sum_{j\in J^J_m}\alpha_{j,t}}_{\text{Joining MPs}} \tag{S5}$$

Here, $\mu_{0,t-1}$ is a weighted average of the finite-sample average of the "remainers" and "leavers" in $t-1$, where the weights are given by the relative proportion of those groups in that parliamentary session. $\mu_{0,t}$ is constituted from the equivalent averages for "remainers" and "joiners" in time period t, again weighted by the size of those two groups in t.

Taking the difference between S4 and S5 and rearranging reveals an additive decomposition which separates the two effects of interest:

$$\begin{aligned}\mu^m_{0,t} - \mu^m_{0,t-1} = {} & \underbrace{\pi^R_{t}\frac{1}{|J^R_m|}\sum_{j\in J^R_m}\alpha_{j,t} - \pi^R_{t-1}\frac{1}{|J^R_m|}\sum_{j\in J^R_m}\alpha_{j,t-1}}_{\text{"Within-MP" effect } (S_m)} + \\ & \underbrace{(1-\pi^R_{t})\frac{1}{|J^J_m|}\sum_{j\in J^J_m}\alpha_{j,t} - (1-\pi^R_{t-1})\frac{1}{|J^L_m|}\sum_{j\in J^L_m}\alpha_{j,t-1}}_{\text{"Replacement" effect } (R_m)}\end{aligned} \tag{S6}$$

We denote the within-MP effect for men as W_m and the replacement effect as R_m. We

can also, of course, define the same quantities for female MPs, and therefore can describe the changing gender difference in terms of replacement and socialisation effects:

$$(\mu^{w}_{0,t} - \mu^{m}_{0,t}) - (\mu^{w}_{0,t-1} - \mu^{m}_{0,t-1}) = \underbrace{(W_w - W_m)}_{\text{“Within-MP” difference}} - \underbrace{(R_w - R_m)}_{\text{“Replacement” difference}} \tag{S7}$$

Turning to our results, we plot these quantities in the left (for male MPs) and centre (for female MPs) panels of figure S10. The x-axis describes the average direction and magnitude of changes between parliamentary sessions for each style for men and women, respectively. The right-hand panel reports *the difference* in the effects for women and men. In each panel, hollow points show changes that occur because of replacement, and solid points show changes that occur due to within-MP shifts.

We use these plots to understand whether replacement or within-MP change is a stronger determinant of the aggregate shifts we observe. Overall, neither differential replacement nor within-MP change alone explain the convergence that we document across multiple different styles in the main body of the paper, though there is some evidence that replacement is more important as a mechanism for explaining the changing gender dynamics we observe for the “agentic” styles while within-MP change is somewhat more important for explaining change for more “communal” styles.

For example, figure 2 in the main body of the paper shows that women are much more likely than men to use negative emotion in their speeches in later years, but only somewhat more likely in the earlier years. The middle panel of figure S10 shows that the replacement effect for women for negative emotion is positive (the hollow point for negative emotion is greater than zero), which implies that newly elected women are more negative than the women leaving parliament, on average. However, the left panel of figure S10 suggests that this is *not* true for male MPs: male MPs joining parliament use negative language at the same rate on average as male MPs leaving parliament (the

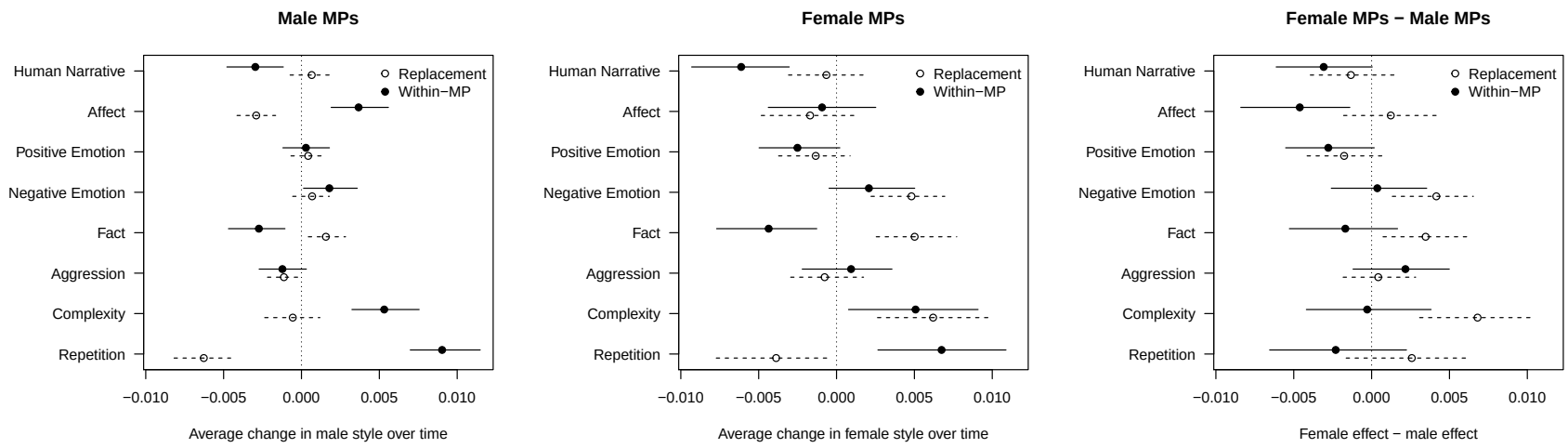

Figure S10: Within-MP and replacement over time change by gender

hollow point for negative emotion is close to zero). As a consequence, the right panel suggests a (positive) differential replacement effect for negative emotion. Note that the difference between male and female *within-MP* effects is close to zero for negative emotion. This suggests that the divergence between men and women that we note at the aggregate level is almost entirely driven by differential replacement between male and female MPs, rather than existing MPs becoming more alike in their behaviour over time.

The right-hand panel of S10 indicates that, beyond negative emotion, replacement effects also account for a greater share of the aggregate change in gender differences for factual language and complexity. For both, while the women entering parliament are significantly more likely to use these styles than the women leaving, newly elected male MPs employ these styles at broadly similar rates as the men that they replace. By contrast, both male *and* female MPs are less likely to use factual language as their careers in parliament progress, and the speeches of both men *and* women become more complex the longer they spend in parliament. As a consequence, the large aggregate shifts that we observe for these styles are largely driven by the fact that the women newly elected to parliament adopted a legislative debating style that was more factual, complex, and negative than the women they replaced.

For other styles, we see that within-MP change accounts for a greater share of the variation in gender differences. For instance, the gradually decreasing gap in the use of human narrative in figure 2 in the main body of the paper is mostly attributable to women MPs using this style less the longer that they stay in parliament, but the decreasing use of human narrative for male MPs is much smaller. Similarly, on average women employ less positive emotion over time, whereas the positive language use of male MPs remains relatively constant. Conversely, within-MP change in affect for men is positive, implying that men become more emotional overall in their speeches over time, but there is very

little average within-MP change in affect for women. These results imply that, for these style types, the convergence that we see in the main analysis is driven by the different stylistic trajectories than male and female MPs appear to follow throughout their tenure in parliament.

Topic-based confounding

We present evidence of convergence between men and women with respect to several debating styles over time. One potential concern for the interpretation of our results is that the parliamentary agenda is not fixed, and changes to the set of issues under discussion may result in convergence between men and women even in the absence of behaviour change.

Consider, for instance, a style like human narrative, where we observe a large convergence between men and women over time. Women are significantly more likely to use human narrative in their parliamentary speeches at the beginning of the time period than they are at the end. If, however, women are more likely to use human narrative than men in certain *topics*, and those topics become less prevalent over time, then the convergence we document might in fact be attributable to changes to the parliamentary agenda. For changes in topic prevalence to be responsible for convergence, it would have to be the case that the topics on which we observe women using *more* human narrative than men are becoming *less* prevalent, or that the topics on which we see women using *less* human narrative than men are become *more* prevalent over time. For example, perhaps women use more human narrative than men when discussing education policy, and education policy is more frequently discussed in the early period in our data than the later period in our data. If this were true, then our results might be subject to topical confounding, as changes in topical prevalence over time would account for the aggregate changes we observe in the main analysis.

To address this concern, in this section we use statistical topic models to evaluate whether topics on which we observe notable stylistic differences between men and women become more or less prevalent over time. We begin by estimating a correlated topic model for all speeches in our data. As with all topic models, the CTM is an unsu-

pervised learning approach which makes the assumption that the frequency with which words co-occur within different speeches provides information about the topics that feature in those speeches. The CTM requires the analyst to choose the number of topics, K, and here we present results from a $K = 40$ topic model. We implement the CTM as the null form of the Structural Topic Model, which we implement in R (Roberts et al., 2014).

The key output of the topic model is θ, a $N * D$ matrix of topic proportions that measures the degree to which each speech (i) in the data features each of the estimated topics (d). $\theta_{i,k}$ therefore gives the proportion of speech i devoted to topic d. With these topics in hand, we then evaluate – for each of our 8 styles – the size of the stylistic gender gap between men and women on each topic. To do so, we estimate models where we interact the gender of the MP delivering a speech with the topic proportions that pertain to that speech:

$$y^s_{i(j)} = \alpha + \beta^1 Female_j + \sum_{k=2}^{K} \beta^2_k \theta_{i,k} + \sum_{k=2}^{K} \beta^3_k (Gender_i \cdot \theta_{i,k}) + \epsilon_{i(j)} \tag{S8}$$

We use the coefficients of this model to calculate estimated average differences between men and women on speeches devoted to each topic, which we denote as

$$\delta^s_k = \begin{cases} \beta^1 & \text{if } k = 1 \\ \beta^1 + \beta^3_k & \text{if } k \neq 1 \end{cases} \tag{S9}$$

The average difference in style s between between men and women on speeches that are entirely devoted to topic 1 is given by β^1 (i.e. the baseline), and $\beta^1 + \beta^3_k$ captures the average gender difference in style on speeches entirely devoted to topic k. We denote the gender difference on each topic and style as δ^s_k. This specification allows us to capture the aggregate differences between male and female use of a style on each topic.

Positive values for δ_k^s indicate that women use the style more than men in a given topic, and negative values suggest that women use the style less than men in a given topic.

We then estimate a second set of regression models to capture, for each topic, the relationship between time and topic prevalence. To do so, we first multiply the number of words in each speech by the vector of topic proportions for that speech, giving us the weighted number of words dedicated to a given topic for each speech in the data. We then sum these topic-weighted word counts across all speeches within a given calender month, and use the summed word counts as the dependent variable for regressions of the form:

$$y_t^k = \alpha + \gamma_k YearMon_t + \epsilon_t \tag{S10}$$

Here, y_t^k is the number of words on topic k in time period t, and γ_k captures the linear relationship between time and topic prevalence for topic k. Positive values of γ_k imply that topic k becomes more prevalent in parliamentary debate throughout the study period, and negative values suggest that the topic becomes less prevalent over time.

If the topical confounding argument is correct, then for a style like human narrative – where we observe average convergence between men and women over time – it must be the case that there is a negative relationship between the gender gap on that topic and the relationship between topic and time. That is, topics where women use human narrative more than men (positive coefficient from equation S8) should be becoming less prevalent over time (negative coefficient from equation S10).

The topical-confounding hypothesis implies different relationships between topical gender-gaps and changes in topic prevalence over time for different styles. For instance, for human narrative, our main analysis shows that women are more likely to use this style in the early period of our data and less in the later period. For this style, topical

confounding would occur if topics where women use narrative *more* on average than men (positive δ_k^s from equation S9) became *less* prevalent over time (negative γ_k from equation S10), or the topics where women use narrative *less* than men (negative δ_k^s from equation S9) became *more* prevalent over time (positive γ_k from equation S10). For human narrative, then, the topical-confounding hypothesis implies a negative relationship between the two sets of coefficients.

On the other hand, our aggregate results suggest that women are less aggressive than men in the early period of the data, but are equally as aggressive later in the period. Accordingly, if this convergence can be explained by changes to the topics under discussion, it must be the case that the topics on which women tend to be less aggressive than men (negative δ_k^s from equation S9) become less prevalent over time (negative γ_k from equation S10), or that the topics on which women tend to be more aggressive than men (positive δ_k^s from equation S9) become more prevalent over time (positive γ_k from equation S10). Therefore, for aggression, the topical confounding hypothesis implies a positive relationship between the two sets of coefficients.

Following this logic through all eight style types, the topical-confounding explanation suggests that we should observe a positive relationship between γ_k and δ_k^s for aggression, complexity, fact and negative emotion, and a negative relationship between γ_k and δ_k^s for human narrative, affect, positive emotion, and repetition. In figure S11 we evaluate these expectations by plotting the estimated values of γ_k and δ_k^s against each other for each style. In this plot, each point represents a single topic from our topic model, the x-axis measures the gender gap in the use of a given style (δ_k^s), and the y-axis measures the changing prevalence of the topic over time (γ_k). We also fit a regression line between the sets of coefficients, which is coloured in gray if the slope of the line is associated with a p-value of less than 0.05, and otherwise is coloured in red.

The main implication of this analysis is straightforward: we find very little evidence

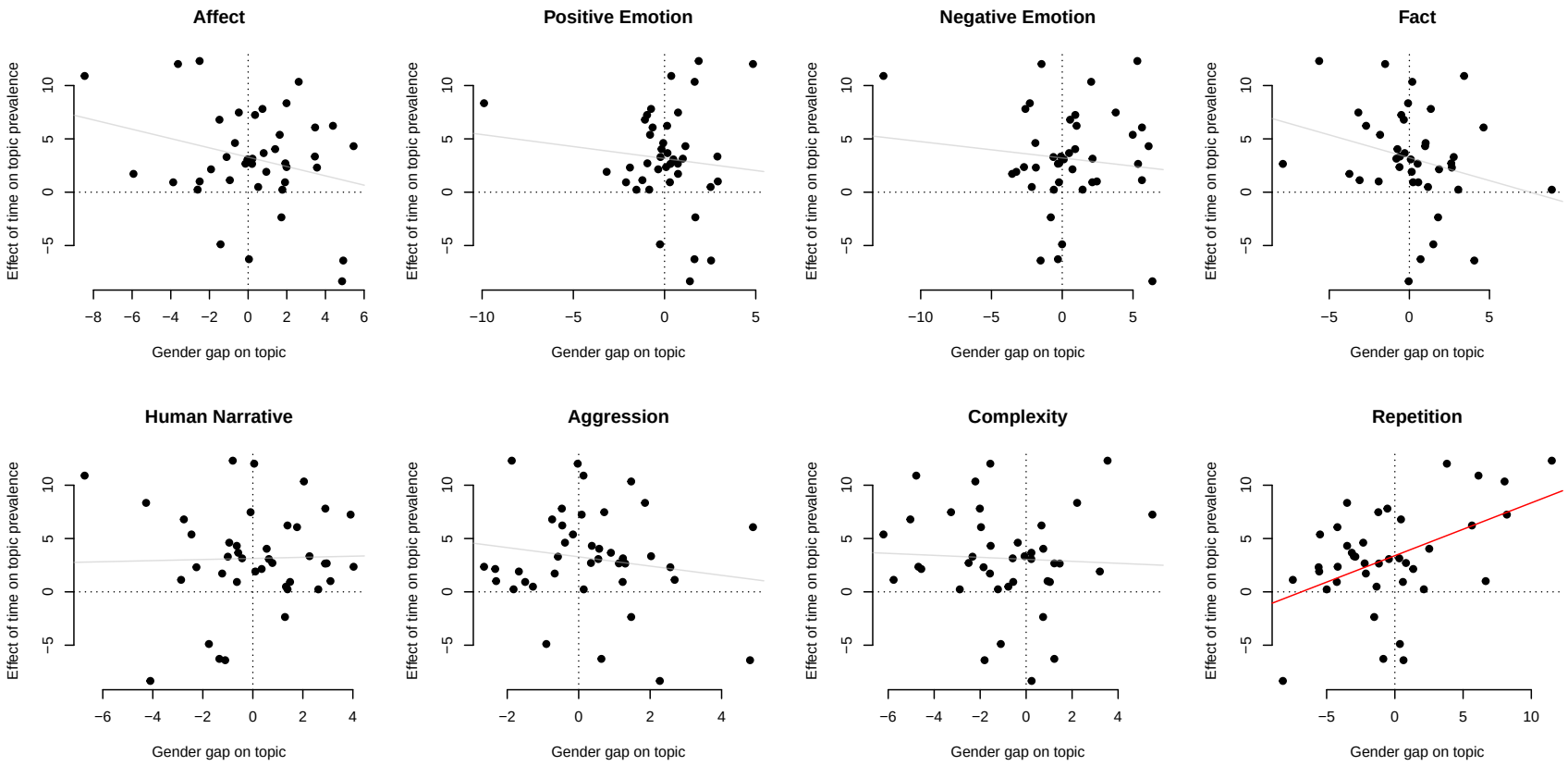

Figure S11: **Topical-confounding**: The figure shows the relationship between the gender gap in the use of a given style on a given topic (x-axis), and the change in the prevalence of a given topic over time (y-axis).

to support the topical-confounding hypothesis. The size of the gender gap measured for a given style on a given topic largely does not predict the degree to which that topic becomes more or less prevalent over time. For three of the styles – aggression, negative emotion and fact – the relationships in figure S11 are negative, where they would need to be positive in order for topical changes to explain the stylistic convergence we document in the main body of the paper. We also find a relationship that is in the "wrong" direction for repetition (that is, although statistically significant, the relationship would need to be negative to cause concern), and there is also essentially no relationship between the gender gap in human narrative on different topics and the changing prevalence of those topics over time. For the remaining styles – affect, positive emotion, and complexity – we do find some evidence that topics on which women display more of these styles become more prevalent over time, but the relationships are very noisy and in none of those cases are we able to reject the null hypothesis of a relationship of zero.

In sum, this analysis implies that the aggregate patterns we observe in the main body of the paper cannot be convincingly explained by changes to the parliamentary agenda over time.

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